

The permutation coboundary constant of the complete complex is $k/3$ for every $k \geq 4$, including the powers of two, with four Hamming-native readings that do not obstruct

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Abstract

Let K_k be the complete two-dimensional complex on k vertices and let $h_1(K_k, \text{Sym})$ be its first coboundary expansion constant with permutation coefficients in the normalised Hamming metric, in the convention that counts cells. Chapman and Lubotzky prove $h_1(K_k, \text{Sym}) \geq k/3$ for every k and every permutation degree. We show that this bound is an equality for every $k \geq 4$, attained by cochains with values in $\text{Sym}(3)$. For $\mathbb{F}_2 = \text{Sym}(2)$ coefficients Kozlov proves the constant is $k/3$ whenever k is not a power of two; it is strictly larger at $k = 4$ (where it is 2) and at $k = 8$ (where it is $20/7$), and at larger powers of two it lies between $k/3$ and $k/3 + 4k/(3(k^2 - 8))$, with strictness open. The permutation constant reaches the bound at every power of two. At $k = 4$ the contrast is sharp: among the subgroups of $\text{Sym}(3)$, the binary subgroup $\mathbb{F}_2$ gives 2, the cyclic subgroup $\mathbb{Z}/3$ gives $3/2$, and only $\text{Sym}(3)$ itself reaches $4/3$. The proof has three parts. At $k = 4, \dots, 8$ an explicit witness is computed in integers, checked by two independent programs, and its gauge minimum is kernel-checked in Lean; at $k = 4$ the minimum also has a short hand proof, and the attaining cochain is a single triangle carrying the three distinct transpositions of $\text{Sym}(3)$. A balanced replication of a cochain multiplies its triangle defect by q^3 and its gauge minimum by exactly q^2 , so the $k = 4$ seed reaches $k/3$ at every multiple of four; the lower bound is proved in full and is an equality of averages, so a cochain attains $k/3$ exactly when every star gauge is optimal. At the remaining k , Kozlov's binary witness embeds through one transposition, and a parity retraction shows that no $\text{Sym}(3)$ gauge does better. The replication law, for every cochain, every k and every q , and the retraction, for every k , are themselves kernel-checked in Lean, so the gauge minimum $6q^2$ of the replicated seed is a kernel-checked fact at every multiple of four. The same value holds in Chapman–Lubotzky's cross-degree convention. The all- k theorem is proved, not searched for: computation supplies explicit witnesses only at $k \leq 8$, and the proof does not depend on the open question of whether $\mathbb{F}_2$ reaches $k/3$ at the larger powers of two. Four further readings of the same metric are recorded because their content is exact: a regularised twisted torsion that is Lipschitz in Hamming distance on the padded family; the displacement Gram of a permutation, which is half the Laplacian of its cycle graph, sums over

a family to half the Laplacian of the Schreier graph of the action, and has a fixed-scale entropy that vanishes exactly on involutions and is bounded by $h(d_H/2)$ for every permutation, with a finite-size deficit given exactly by a lattice sum whose ℓ^3 -scaled limits are $2\zeta(3)$ on even cycles and $\zeta(3)/4$ on odd ones, with every further correction an odd zeta value; an exact embedding of the Hamming metric as a squared distance between graph-subspace projectors, equivalent with explicit constants to Grassmannian geodesic distance; and a fixed-probe response that reads the spectral measure seen by the probe and so separates isospectral defects. None of them obstructs approximation. Grades: [T] derived, [V] computed, [L] kernel-checked in Lean, [C] cited.

1 Setting and conventions

Two permutations of n points are compared by the fraction of points on which they differ, $d_H(\sigma, \tau) = \frac{1}{n} \#\{i : \sigma(i) \neq \tau(i)\}$, a bi-invariant metric on $\text{Sym}(n)$. Following Chapman–Lubotzky [1, 2], a 1-cochain α on a finite 2-complex X assigns a permutation to each oriented edge with $\alpha(v, u) = \alpha(u, v)^{-1}$; its coboundary on a triangle is the holonomy $\delta\alpha(u, v, w) = \alpha(u, v)\alpha(v, w)\alpha(w, u)$; $\|\delta\alpha\|$ is the mean over triangles of $d_H(\delta\alpha(\Delta), \text{Id})$; and $d(\alpha, B^1)$ is the least mean over edges of the distance to a coboundary $\delta\beta(u, v) = \beta(u)^{-1}\beta(v)$. The first coboundary Cheeger constant is

$$h_1^{\text{norm}}(X, \text{Sym}) = \min_{\alpha \notin B^1} \frac{\|\delta\alpha\|}{d(\alpha, B^1)}, \quad h_1^{\text{count}}(X, \text{Sym}) = \frac{|T|}{|E|} h_1^{\text{norm}}(X, \text{Sym}),$$

the first with uniform probability on cells, the second counting cells. On the complete complex $|T|/|E| = (k-2)/3$, so $k/(k-2)$ in the first convention is $k/3$ in the second. The tables use the counting convention, which is the one the enumerations compute and the one in which the $\mathbb{F}_2$ literature states its results. Chapman–Lubotzky define the constant with distance to the cocycles Z^1 ; on the complete complex H^1 vanishes and the two agree [2, Cor. 4.1]. For a cochain α we write $N(\alpha)$ for the total triangle defect and $D(\alpha)$ for the gauge minimum, both counted in moved points; with values in $\text{Sym}(3)$ these are integers and $h_1^{\text{count}} = \min N/D$ over non-coboundaries.

Since $\text{Sym}(2) \cong \mathbb{F}_2$ and d_H on two points is the discrete metric, the $\mathbb{F}_2$ coboundary expansion constant of Linial–Meshulam [3] is the $\text{Sym}(2)$ column of the permutation theory, and $h_1(X, \text{Sym}) \leq h_1(X, \mathbb{F}_2)$ always [2, footnote 2]. Two conventions for Sym are in use, and they are kept apart here. In the *same-degree* convention a cochain and the coboundaries it is compared with all take values in one $\text{Sym}(n)$; write h_1^{same} . In Chapman–Lubotzky’s convention Sym is the disjoint union of all $\text{Sym}(m)$, and a permutation σ of degree n is compared with τ of degree m by the distance with errors

$$d(\sigma, \tau) = 1 - \frac{\#\{j < \min(n, m) : \sigma(j) = \tau(j)\}}{\max(n, m)},$$

which is d_H when $n = m$; write h_1^{err} , where the comparison coboundary may have any degree. Sections 2–4 work in h_1^{same} , and Proposition 11 shows that both constants are $k/3$ on K_k . Where the convention does not matter we write $h_1(K_k, \text{Sym})$.

The question comes from an earlier attempt to extract an obstruction to soficity from operator data after Astra’s non-sofic group of 1 August 2026 [17]. There the Exel–Loring winding of an almost-commuting pair [18] was gated numerically and found to be integral and rigid, but rigid in the operator norm: changing a permutation on εn points costs ε in normalised Hamming distance but an amount of order one in operator norm (2 for a transposition, $\sqrt{3}$ for a 3-cycle), so that

winding jumps under exactly the local surgery soficity forgives. The lesson was to stop fighting the metric from outside and to study quantities native to it. The coboundary constant of the complete complex is the first calibration in that programme, and Section 5 records what the other native readings do.

2 The bound

Proposition 1 (Chapman–Lubotzky [2, Prop. 3.2]; T). *For every 1-cochain α on K_k with coefficients in any group carrying a bi-invariant metric,*

$$\frac{1}{k} \sum_x \|\delta\beta_x - \alpha\| = \frac{k-2}{k} \|\delta\alpha\|,$$

where β_x is the star gauge at x defined below and $\|\delta\beta_x - \alpha\|$ is the mean over edges of $d_H(\alpha(e), \delta\beta_x(e))$. Hence $d(\alpha, B^1) \leq \frac{k-2}{k} \|\delta\alpha\|$, in counting variables $D \leq 3N/k$, and $h_1^{\text{count}}(K_k, \text{Sym}) \geq k/3$ for every k and every permutation degree.

Proof. Fix a vertex x and put $\beta_x(x) = \text{Id}$, $\beta_x(u) = \alpha(x, u)$ for $u \neq x$. On a star edge, $\delta\beta_x(x, u) = \beta_x(x)^{-1}\beta_x(u) = \alpha(x, u)$, so the star edges cost nothing. On an edge (u, v) with $u, v \neq x$, bi-invariance and $\alpha(x, v)^{-1} = \alpha(v, x)$ give

$$d_H(\alpha(u, v), \alpha(x, u)^{-1}\alpha(x, v)) = d_H(\alpha(x, u)\alpha(u, v)\alpha(v, x), \text{Id}) = d_H(\delta\alpha(x, u, v), \text{Id}).$$

So after the star gauge at x each non-star edge carries exactly the holonomy of the one triangle it spans with x . The cost of a holonomy does not depend on the base point or the orientation, because changing either replaces it by a conjugate or an inverse and $d_H(\cdot, \text{Id})$ is invariant under both. Summing over x , every triangle is counted once for each of its three vertices:

$$\sum_x \|\delta\beta_x - \alpha\| = \frac{1}{|E|} \sum_x \sum_{\substack{u < v \\ u, v \neq x}} d_H(\delta\alpha(x, u, v), \text{Id}) = \frac{3|T|}{|E|} \|\delta\alpha\| = (k-2) \|\delta\alpha\|,$$

using $|T|/|E| = (k-2)/3$. Dividing by k gives the identity. The least of the k star costs is at most their mean, so $d(\alpha, B^1) \leq \frac{k-2}{k} \|\delta\alpha\|$; in moved points the same count reads $\sum_x \text{cost}(\beta_x) = 3N$ and $D \leq 3N/k$. $\square$

The identity is an equality of averages, not an estimate, and it also says when the bound is met.

Corollary 2 (The equality case; T). *A cochain $\alpha \notin B^1$ has $N/D = k/3$ if and only if every star gauge β_x is an optimal gauge for α .*

Proof. Each star cost is at least D and their mean is $3N/k$. So $D = 3N/k$ holds exactly when no star cost exceeds D . $\square$

Every witness of Sections 3 and 4 is therefore a cochain all of whose star gauges are optimal. Script K3-V4-STAR checks this on every witness of Appendix A.

Remark 3 (Bi-invariance is the hypothesis; V). Replacing d_H by a site-weighted Hamming metric, still a metric but no longer bi-invariant, moves the ratio of Proposition 1 off $(k-2)/k$ at once: at $k = 4$ the weighted ratio ranges over 0.486–0.548 against an exact 0.500, with the same drift at $k = 5, 6$.

3 The bound is attained at $k = 4, \dots, 8$: computed witnesses

Theorem 4 (T+V+L). *For $k = 4, 5, 6, 7, 8$, $h_1^{\text{count}}(K_k, \text{Sym}) = k/3$ and $h_1^{\text{norm}}(K_k, \text{Sym}) = k/(k-2)$, attained by a cochain with values in $\text{Sym}(3)$. The values of the coefficient subgroups are*

h_1^{count}	$k = 4$	$k = 5$	$k = 6$	$k = 7$	$k = 8$
$\text{Sym}(2) = \mathbb{F}_2$	2	5/3	2	7/3	20/7
$\text{Sym}(3)$	4/3	5/3	2	7/3	8/3
lower bound $k/3$	4/3	5/3	2	7/3	8/3

The $\mathbb{F}_2$ row is Meshulam–Wallach and Kozlov [4, 5], including Kozlov’s value $8/3 + 4/21 = 20/7$ at $k = 8$; the $\text{Sym}(3)$ row is this paper.

Proof. The lower bound is Proposition 1. For the upper bound one cochain per k suffices, provided its ratio is computed exactly and its gauge minimum is the true minimum over all gauges. With $\text{Sym}(3)$ coefficients every Hamming distance is 0, $2/3$ or 1, so all quantities are moved-point counts in integers and “ $= k/3$ ” is rational equality. The gauge minimum is taken over all 6^{k-1} gauges with $\beta(\text{root}) = \text{Id}$, which loses nothing because a constant right factor conjugates every edge term and the moved-point count is conjugation-invariant. The witnesses are listed in Appendix A; their triangle defects and gauge minima are

k	4	5	6	7	8
N/D	8/6	10/6	36/18	28/12	72/27

and each equals $k/3$. Every witness was recomputed by a second program sharing no code with the search (Appendix B), which returned the same N , the same D and the same minimising gauge. The same five equalities are kernel-checked in Lean (Section 6). $\square$

3.1 The four-vertex seed has a hand proof

Proposition 5 ($k = 4$; T+V). *Let α be the identity on the three edges at vertex 0 and carry the three distinct transpositions a, b, c of $\text{Sym}(3)$ on the edges 12, 13, 23. Then $N(\alpha) = 8$ and $D(\alpha) = 6$, so $N/D = 4/3 = k/3$. Over all $\text{Sym}(3)$ -cochains on K_4 the minimum ratio is $4/3$; restricted to $\text{Sym}(2)$ it is 2, and restricted to $\mathbb{Z}/3$ it is $3/2$.*

Proof. All four triangle holonomies are transpositions, so $N = 8$; the identity gauge costs 6. Suppose a gauge cost less than 6. Every non-identity residual edge costs at least 2, so at most two edges are non-identity. Every triangle has odd holonomy, so the residual must place a transposition on every triangle: a single transposition meets only two of the four triangles, and two transpositions sharing a vertex leave the opposite triangle even. Hence the residual is two transpositions on a perfect matching, with no budget for a 3-cycle. But then the two triangles containing the matching edge at vertex 0 have equal holonomy based at 0, whereas the three triangle holonomies of α based at 0 are a, b, c , all distinct, and a gauge conjugates all three by the same permutation, which preserves distinctness. So $D = 6$. The minimum $4/3$ over all $\text{Sym}(3)$ -cochains then needs no search: Proposition 1 bounds every ratio below by $4/3$, and α attains it. Only the subgroup minima, 2 for $\text{Sym}(2)$ and $3/2$ for $\mathbb{Z}/3$, come from exhausting all 6^6 cochains on K_4 [V]. $\square$

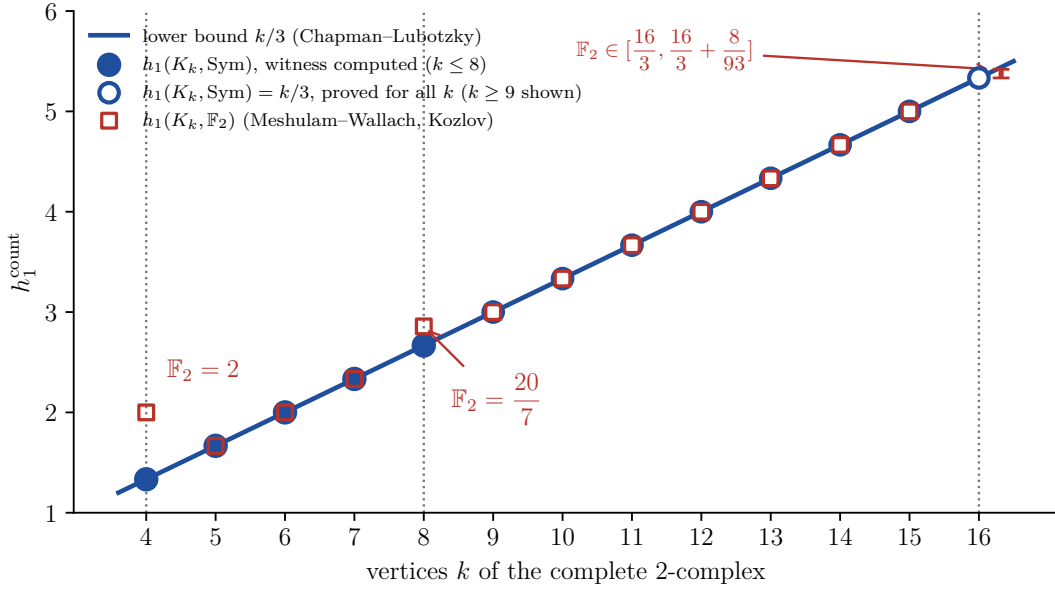


Figure 1: The bound and the constant. The permutation constant sits on $k/3$ at every k : computed and kernel-checked for $k \leq 8$ (filled), proved by Theorem 8 for every k (open). The $\mathbb{F}_2$ constant is above the bound at $k = 4$ (2) and $k = 8$ ($20/7$); at $k = 16$ Kozlov’s bounds confine it to the bracketed interval $[16/3, 16/3 + 8/93]$, and whether it lies strictly above $16/3$ is open.

The mechanism is non-abelian. The product of the three distinct transpositions around the triangle is a transposition, while each adjacent product is a 3-cycle (kernel-checked as `k4_mechanism`). A single transposition on one edge ($\mathbb{F}_2$) pays 2 on each of the two triangles through it against a gauge cost of 2, ratio 2; two 3-cycles pay 3 on two triangles against gauge cost 4, ratio $3/2$. Only three *different* involutions reach the bound.

3.2 The other computed witnesses

At $k = 5, 6, 7$ the $\mathbb{F}_2$ theory already reaches $k/3$ by Kozlov’s theorem, and the witnesses found are the $\mathbb{F}_2$ ones at $k = 5$ (a path on four vertices) and $k = 7$ (a four-cycle with two pendant edges), both carrying a single transposition. At $k = 6$ the witness found gauges to six 3-cycles, a $\mathbb{Z}/3$ -valued cochain. At $k = 8$ the witness, gauged to its minimising coboundary, has residual support on 13 of the 28 edges: twelve transpositions, using all three transpositions of $\text{Sym}(3)$, and one 3-cycle (Figure 3). A cochain valued in a single transposition is an $\mathbb{F}_2$ cochain, and $\mathbb{F}_2$ cannot reach $8/3$ at $k = 8$, so this witness must use more of $\text{Sym}(3)$. Section 4 gives a second, simpler $k = 8$ witness with $(N, D) = (64, 24)$ and twelve transposition edges.

3.3 What the search is and is not

The witnesses at $k = 5, \dots, 8$ were found by simulated annealing over cochains, with the exhaustive gauge minimum recomputed exactly at every step (the 6^{k-1} -entry cost vector is updated incrementally when one edge changes). A search that returns $k/3$ is a proof, because an upper-bound witness has met a proven lower bound. A search that does not is nothing: at $k = 8$ one annealing seed reached $8/3$ at restart 13 (counting from 0) while a second stalled at $87/32$

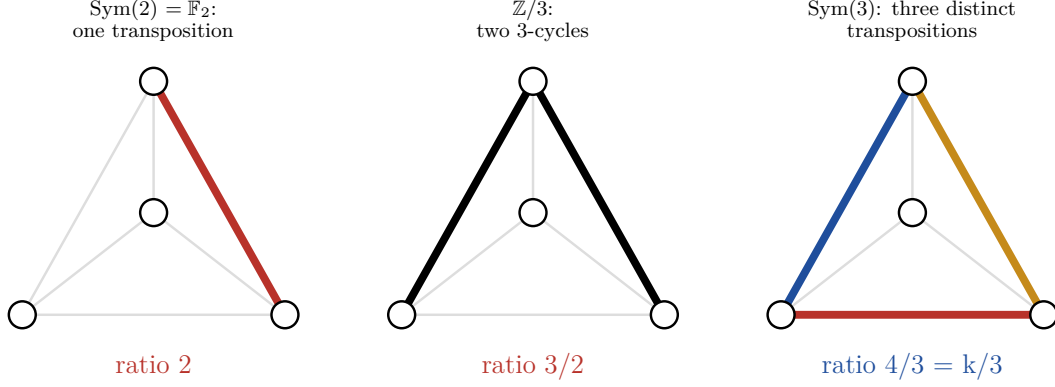


Figure 2: $k = 4$ by coefficient subgroup. The bound needs the non-abelian group, not a larger abelian one.

through twenty-two restarts, and earlier hill climbs without incremental updates stalled at 31/11. Exhaustion over cochains is available only at $k = 4$: K_5 already carries $6^{10} \approx 6.0 \times 10^7$ cochains and K_6 carries $6^{15} \approx 4.7 \times 10^{11}$, while K_{16} has 6^{15} gauges for each cochain. Past $k = 8$ the proof proceeds by construction.

4 The bound is attained for every $k \geq 4$

Lemma 6 (Balanced replication; T+L). *Replace each vertex i of K_k by a cluster of $q \geq 1$ vertices (i, a) . Put the identity on every edge inside a cluster and α_{ij} on every edge from cluster i to cluster j ; call the resulting cochain $\alpha^{[q]}$ on K_{qk} . Then, for any coefficient group with a bi-invariant metric,*

$$N(\alpha^{[q]}) = q^3 N(\alpha), \quad D(\alpha^{[q]}) = q^2 D(\alpha), \quad \text{so} \quad \frac{N(\alpha^{[q]})}{D(\alpha^{[q]})} = q \frac{N(\alpha)}{D(\alpha)} \quad (\alpha \notin B^1).$$

This is an exact scaling law for every cochain, not only for extremisers.

Proof. A triangle with vertices in three distinct clusters carries the holonomy of the underlying triangle of K_k , and each triangle of K_k has q^3 such lifts. A triangle with two vertices in one cluster carries $\text{Id} \cdot \alpha_{ij} \cdot \alpha_{ij}^{-1} = \text{Id}$, and a triangle inside one cluster carries Id . This gives the numerator.

For the denominator take an arbitrary gauge $\beta_{(i,a)}$ on K_{qk} , with no assumption that it is constant on clusters. A *section* is a choice of one representative $A_i \in \{1, \dots, q\}$ in each cluster; there are q^k of them. Each section gives a gauge $\gamma_i^A = \beta_{(i,A_i)}$ on K_k , whose cost is at least $D(\alpha)$ by definition of D . An inter-cluster edge $((i, a), (j, b))$ is used by exactly the sections with $A_i = a$ and $A_j = b$, and there are q^{k-2} of those. Summing the costs of all sections therefore counts every inter-cluster edge of K_{qk} exactly q^{k-2} times:

$$q^k D(\alpha) \leq \sum_A \text{cost}(\gamma^A) = q^{k-2} \sum_{i < j} \sum_{a, b} d_H(\beta_{(i,a)}^{-1} \alpha_{ij} \beta_{(j,b)}, \text{Id}) = q^{k-2} \text{cost}_{\text{inter}}(\beta).$$

So the inter-cluster cost of β is at least $q^2 D(\alpha)$, and since the intra-cluster cost is non-negative, every gauge costs at least $q^2 D(\alpha)$. This is a finite double count; no probability and no averaging

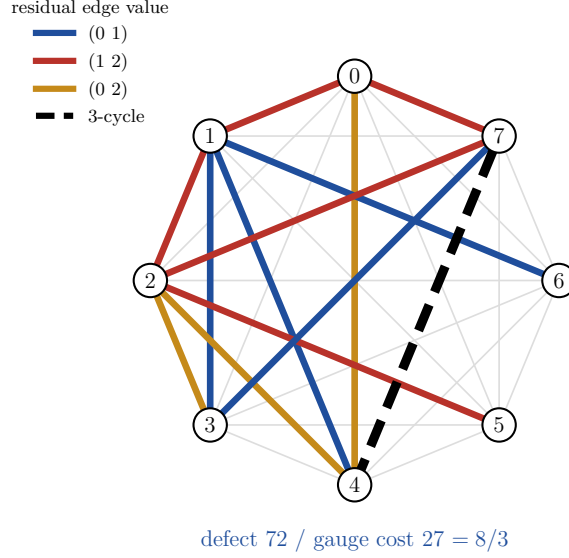


Figure 3: The $(72, 27)$ witness at $k = 8$ after gauging: 13 of 28 edges non-identity, grey edges the identity.

over an infinite set is used. Conversely, copying an optimal gauge of α across each cluster costs nothing inside clusters and exactly $q^2 D(\alpha)$ between them. For an infinite coefficient group, use near-optimal gauges and take the infimum. $\square$

Corollary 7 (Every multiple of four; T+V+L). *For $k = 4q$ the replicated seed of Proposition 5 has $N = 8q^3$ and $D = 6q^2$, so $N/D = 4q/3 = k/3$. It has exactly $3q^2$ non-identity edges, all transpositions, and uses only permutation degree three.*

At $q = 2$ the replicated K_8 has $(N, D) = (64, 24)$, confirmed by exhausting all 6^7 root-fixed gauges. For every q the lemma and the corollary are kernel-checked (`rep_defect`, `rep_attain`, `rep_lower`, `rep_seed`; Section 6), the lower bound through the same count over the q^k sections, with no assumption that the gauge is constant on clusters. Beyond that the minima are values of the lemma, not enumerations: the defects counted explicitly are $N = 216, 512, 4096, 32768$ at $k = 12, 16, 32, 64$, against lemma minima $D = 54, 96, 384, 1536$.

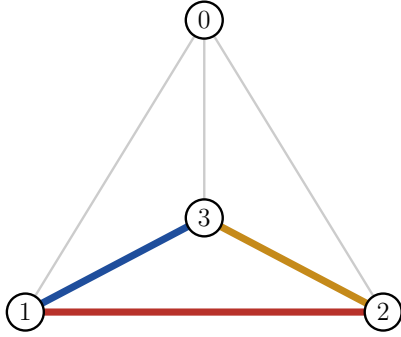
Theorem 8 (T). *For every $k \geq 4$,*

$$h_1^{\text{count}}(K_k, \text{Sym}) = \frac{k}{3}, \quad h_1^{\text{norm}}(K_k, \text{Sym}) = \frac{k}{k-2},$$

and the minimum is attained by a cochain with values in $\text{Sym}(3)$.

Proof. The lower bound is Proposition 1. If k is a multiple of four, Corollary 7 gives a witness; this covers every power of two ≥ 4 . Otherwise k is not a power of two and Kozlov proves $h_1^{\text{count}}(K_k, \mathbb{F}_2) = k/3$ [5, Thm. 4.2, Cor. 4.3]. Embed the binary group as $H = \{\text{Id}, t\}$ for one transposition t of $\text{Sym}(3)$ and read Kozlov's witness as an H -valued cochain. Counted in moved points, numerator and denominator are both twice their binary values, since a transposition moves two points (in normalised units the common factor is $2/3$); either way the ratio is unchanged

seed K_4 :
 $N = 8, D = 6$



replicated K_8 ($q = 2$):
 $N = 8q^3 = 64, D = 6q^2 = 24$

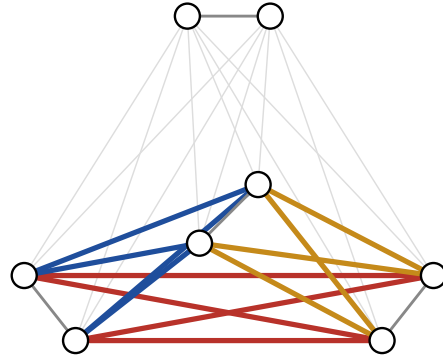


Figure 4: Balanced replication. Each vertex of the $k = 4$ seed becomes a cluster of q vertices joined by identity edges (grey); each coloured seed edge becomes the q^2 edges between two clusters. Triangles across three clusters copy the seed holonomy, triangles with two vertices in one cluster close to the identity, and the gauge minimum scales exactly by q^2 .

when gauges range over H . Gauges outside H do no better. The retraction $r : \text{Sym}(3) \rightarrow H$, $r(g) = t^{\text{parity}(g)}$, is a homomorphism that restricts to the identity on H , and it does not increase the moved-point count: an even element goes to Id at cost 0, and an odd element is a transposition of cost 2 and goes to t at cost 2. For an H -valued cochain α and any $\text{Sym}(3)$ -gauge β , the edge term $r(\beta_u^{-1} \alpha_{uv} \beta_v) = r(\beta_u)^{-1} \alpha_{uv} r(\beta_v)$ is the edge term of the H -gauge $r \circ \beta$ and costs no more. So the $\text{Sym}(3)$ gauge minimum equals the H gauge minimum, and the ratio is $k/3$. $\square$

Remark 9 (What each part rests on). The lower bound $k/3$ is Chapman–Lubotzky’s star average [2, Prop. 3.2], proved in Section 2 [T]; it is not formalised for general k . Equality at $k = 4, \dots, 8$ is by explicit $\text{Sym}(3)$ witnesses whose gauge minima are kernel-checked [V+L]. Equality at every multiple of four is the balanced replication of the $k = 4$ seed, kernel-checked for every q [T+L]. Equality at the remaining k is Kozlov’s binary witness [C] together with the parity retraction, which is kernel-checked for every k [T+L]. Kozlov’s theorem and the assembly of the three cases into Theorem 8 are not in the Lean library.

For comparison, Kozlov’s results at the powers of two are exact at the two smallest cases, $h_1^{\text{count}}(K_4, \mathbb{F}_2) = 2$ and $h_1^{\text{count}}(K_8, \mathbb{F}_2) = 20/7$, and for $k = 2^d$ with $d \geq 3$ the doubled staircase gives

$$\frac{k}{3} \leq h_1^{\text{count}}(K_k, \mathbb{F}_2) \leq \frac{k}{3} + \frac{4k}{3(k^2 - 8)}$$

[5, Thm. 4.6], so $16/3 + 8/93$ at $k = 16$ and $32/3 + 16/381$ at $k = 32$. Whether the excess is strict at every power of two is open, and nothing above uses it.

Remark 10 (Which coefficient subgroup is needed; T+C). For k not a power of two the binary subgroup $\{\text{Id}, t\}$ already attains $k/3$ (Kozlov), and the $k = 6$ witness of Section 3 shows that $\mathbb{Z}/3$ can attain it too. At $k = 4$ neither abelian subgroup does: $\mathbb{F}_2$ gives 2 and $\mathbb{Z}/3$ gives $3/2$ (Proposition 5). At $k = 8$, $\mathbb{F}_2$ gives $20/7 > 8/3$. At the larger powers of two, whether $\mathbb{F}_2$ reaches

$k/3$ is Kozlov's open question; the proof above does not need it, because the non-abelian seed is replicated. So non-abelian coefficients are what reach the bound at every power of two, and they are proved necessary at $k = 4$ and against $\mathbb{F}_2$ at $k = 8$. In terms of permutation degree: at $k = 4$ and $k = 8$, degree 3 is the smallest degree at which the bound is attained, since degree 2 is $\mathbb{F}_2$. At larger powers of two degree 3 suffices, and whether degree 2 also suffices is exactly Kozlov's open binary question.

The blow-up is adjacent to the multi-edge blow-ups of Dikstein and Dinur [6], which reduce coboundary expansion of complexes with multiple edges to their flattenings in a different setting; the exact scaling $N \mapsto q^3 N$, $D \mapsto q^2 D$ used here does not appear there as stated.

4.1 The cross-degree convention

Proposition 11 (T+V+C). *For every $k \geq 4$, $h_1^{\text{err,count}}(K_k, \text{Sym}) = h_1^{\text{same,count}}(K_k, \text{Sym}) = k/3$.*

Proof. Throughout, the cost of a comparison is the edge-distance sum $\sum_e d(\alpha(e), \delta\beta(e))$, with d the distance with errors of Section 1.

Lower bound. For a cochain of degree n , the star gauge of Proposition 1 is a coboundary of the same degree, and between permutations of equal degree the distance with errors is d_H . So the star-gauge identity bounds the cross-degree distance to B^1 exactly as before, and $h_1^{\text{err,count}} \geq k/3$. (This is Chapman–Lubotzky's own setting [1, 2].)

Upper bound: what has to be shown. The witnesses of Theorem 8 have degree three and attain $k/3$ against degree-three coboundaries. It remains to show that no comparison coboundary of another degree m repairs them more cheaply. Two baselines do most of the work. A degree-three permutation and a degree-2 permutation agree on at most two of the three points, so every edge costs at least $1/3$. A degree-three permutation and one of degree $m \geq 4$ agree on at most three of m points, so every edge costs at least $1 - 3/m \geq 1/4$.

The seed. On K_4 the seed's degree-three repair cost is $D = 6$ moved points, which is 2 in edge-distance units. For $m \geq 5$ the baseline alone is $6(1 - 3/m) \geq 12/5 > 2$. For $m = 2, 3, 4$ exact enumeration over all comparison coboundaries gives minima $10/3$, 2 and 3 [V]. So the seed's cross-degree minimum is 2.

Replicated seeds. The section count in Lemma 6 uses nothing about the degree of β : each section is a degree- m gauge on K_4 , whose cross-degree cost is at least the seed's minimum. So no replicated seed has a cheaper cross-degree repair.

Embedded binary witnesses. Write $k = q(2t + 1)$ with $t \geq 1$, which is possible exactly when k is not a power of two. Kozlov's witness is the q -fold blow-up of the staircase on $2t + 1$ vertices, with binary support of $D_2 = q^2 t(t + 1)/2$ edges [5, Thm. 4.2]. In $\text{Sym}(3)$ each support edge carries a transposition at distance $2/3$, so its degree-three repair cost is $D_3 = \frac{2}{3} D_2 = q^2 t(t + 1)/3$. A degree-2 comparison costs at least $|E|/3$ and a degree- $m \geq 4$ comparison at least $|E|/4$, with $|E| = k(k - 1)/2$. Since $|E|/3 \geq |E|/4$, it suffices that $|E|/4 \geq D_3$, that is,

$$\begin{aligned} \frac{q(2t + 1)(q(2t + 1) - 1)}{8} &\geq \frac{q^2 t(t + 1)}{3} \iff 3q(2t + 1)^2 - 3(2t + 1) \geq 8q t(t + 1) \\ &\iff q(4t^2 + 4t + 3) \geq 6t + 3, \end{aligned}$$

which holds for all $q, t \geq 1$ because $4t^2 + 4t + 3 > 6t + 3$. So no other degree repairs these witnesses more cheaply either.

Summary of the cases. Every witness used in Theorem 8, against every comparison degree m , is covered by exactly one of the arguments above:

witness	$m = 2$	$m = 3$	$m \geq 4$
seed on K_4	enumeration: $10/3 > 2$	Proposition 5: 2	$m = 4$: enumeration, $3 > 2$; $m \geq 5$: baseline $\geq 12/5 > 2$
replicated seed, $k = 4q$	section count of Lemma 6, which reduces every m to the seed row		
binary witness, $4 \nmid k$	baseline $ E /3 \geq D_3$	parity retraction (Theorem 8)	baseline $ E /4 \geq D_3$

The lower bound $k/3$ holds for every m by the star-gauge identity, as shown first. $\square$

The support counts match the computed binary witnesses: $k = 5$ ($q = 1, t = 2$) has 3 edges and $k = 7$ ($q = 1, t = 3$) has 6, the path on four vertices and the four-cycle with two pendants of Section 3. At $k = 6$ ($q = 2, t = 1$) Kozlov's witness is the four edges of $K_{2,2}$ between two pairs of vertices, which has 8 odd triangles against 4 edges, ratio $2 = 6/3$. The search found a different minimiser there, a $\mathbb{Z}/3$ -valued cochain.

5 Four readings of the same metric that do not obstruct

This section is a calibration section of negative results. Each reading was built in pursuit of a Hamming-stable global invariant. Each is recorded because its content is exact, and because together they locate what such an invariant would have to be.

5.1 The regularised torsion is Lipschitz on the padded family

Take a two-bridge knot group $\langle a, b \mid waw^{-1}b^{-1} \rangle$, a permutation representation into $\text{Sym}(n_0)$, and Wada's twisted Alexander invariant computed by Fox calculus. For ten knots ($3_1, 4_1, 5_1, 5_2, 6_1, 6_2, 6_3, 7_1, 7_2, 7_4$) the trivial representation returns the classical Alexander polynomial up to a unit with $|c| = 1$ to 10^{-15} [V], and each knot has a $\text{Sym}(3)$ representation whose twisted polynomial is a non-unit [V], the Silver–Williams phenomenon [8]. Perturbing one generator by a transposition destroys the homomorphism, and the natural surrogate is $T_\eta = \frac{1}{2n} \log \det(A^*A + \eta^2 I)$ of the twisted Fox matrix.

Proposition 12 (T+V). *Pad the representation with fixed points to size n . Padding block-diagonalises the Fox matrix, and a transposition inside the padding perturbs one block of size independent of n ; hence the change satisfies $|\Delta T_\eta| = C_{K,\eta} d_H$ exactly, with $C_{K,\eta}$ depending on the knot, the representation, the perturbation and the regulator η , but not on n . Explicitly, if A_0 and A_1 are the affected block before and after the perturbation and*

$$R_\eta = \frac{\det(A_1^* A_1 + \eta^2 I)}{\det(A_0^* A_0 + \eta^2 I)},$$

then $\Delta T_\eta = \frac{1}{2n} \log R_\eta$; a transposition has $d_H = 2/n$, so $C_{K,\eta} = \frac{1}{4} |\log R_\eta|$ is an identity, not a fit. Over sixteenfold growth in n the ratio is flat to every printed digit (for example 0.894 for 3_1 , 0.029 for 4_1 , 1.041 for 5_2), and on this family the measured $C_{K,\eta}$ does not change over $\eta = 10^{-2}$ to 10^{-5} [V]; that is an observation on this family, not a proof that the regulator drops out.

Corollary 13. *On the padded family the change ΔT_η is proportional to the defect and vanishes with it, so it cannot supply a lower bound that survives when the defect fraction tends to zero: it carries no obstruction there. Whether $|T_\eta(\alpha) - T_\eta(\beta)| \leq C d_H(\alpha, \beta)$ for arbitrary approximate representations, and whether the regulator drops out in general, are open; padding alone does not imply either.*

An independent replay is shipped with this version (script K3-V7-TORSION). It first passes a control: with the trivial representation its Fox matrix returns the Alexander polynomials of 3_1 , 4_1 and 5_2 up to a unit. It then uses its own choices, namely $t = 1.07$, the surjective $\text{Sym}(3)$ representation of 3_1 and the abelian one for 4_1 and 5_2 , and the generator a perturbed by a transposition of two padding points. It gives $C_{K,\eta} = 0.07503, 0.06579$ and 0.16247 , flat over $n = 6, \dots, 96$ to every printed digit and equal to $\frac{1}{4}|\log R_\eta|$ exactly, with a relative change of order 10^{-4} over $\eta = 10^{-2}$ to 10^{-5} [V]. Its constants differ from the August values above because the constant depends on these choices, as the proposition says.

The same functional fails to separate arrangement: ten per cent of points moved as disjoint transpositions or as one long cycle change T_η by comparable amounts, with no consistent direction across the ten knots [V]. That is a fact about a scalar, and the next reading says what a scalar discards.

5.2 The displacement Gram is the cycle Laplacian

Let π be a permutation of n points with permutation matrix $Ue_i = e_{\pi(i)}$, and let D be its set of moved points. Put $B = (U - I)/\sqrt{2}$, so that the columns of B indexed by D are the unit displacement vectors $v_i = (e_{\pi(i)} - e_i)/\sqrt{2}$, and let

$$G = B^*B = I - \frac{1}{2}(U + U^*),$$

restricted to the moved coordinates.

Theorem 14 (T+L). $\langle v_i, v_j \rangle = 1$ for $i = j$ and $\langle v_i, v_j \rangle = -\frac{1}{2}([\pi(i) = j] + [\pi(j) = i])$ for $i \neq j$. Hence $G = \frac{1}{2}L$, where $L = 2I - A$ is the Laplacian of the cycle graph of π read as a multigraph, a 2-cycle being a doubled edge. For an ℓ -cycle the eigenvalues are $2\sin^2(\pi j/\ell)$, $0 \leq j < \ell$.

Proof. $\langle e_{\pi(i)} - e_i, e_{\pi(j)} - e_j \rangle = 2[i = j] - [\pi(i) = j] - [\pi(j) = i]$, which is kernel-checked as `disp_gram`. For a transposition both indicators hold and the off-diagonal entry is -1 , which is why C_2 must be the doubled edge. For the eigenvalues, list the points of one ℓ -cycle as $p_0, \dots, p_{\ell-1}$ with $\pi(p_s) = p_{s+1}$, put $\omega = e^{2\pi i/\ell}$ and $w_j = \sum_s \omega^{js} e_{p_s}$. Then $Uw_j = \omega^{-j}w_j$, so $Gw_j = (1 - \frac{1}{2}(\omega^j + \omega^{-j}))w_j = 2\sin^2(\pi j/\ell)w_j$, and the ℓ vectors w_j form a basis. The rewriting $2 - 2\cos 2x = 4\sin^2 x$ is kernel-checked as `cycle_laplacian_eigen`. $\square$

Theorem 15 (The Gram spectrum determines the cycle type; T+V). Let a_ℓ be the number of ℓ -cycles of π and T_r the Chebyshev polynomial of degree r . Then

$$F_r := \text{Tr } T_r(I - G) = \text{Tr } U^r = \sum_{\ell|r} \ell a_\ell, \quad a_\ell = \frac{1}{\ell} \sum_{d|\ell} \mu(\ell/d) F_d,$$

with μ the Möbius function. So the spectrum of G , through the numbers F_r , determines the cycle type, and the formula reconstructs it.

Proof. $I - G = \frac{1}{2}(U + U^*)$ has eigenvalues $\frac{1}{2}(z + z^{-1})$ over the eigenvalues z of U , and $T_r(\frac{1}{2}(z + z^{-1})) = \frac{1}{2}(z^r + z^{-r})$. Summing, $F_r = \frac{1}{2}(\text{Tr } U^r + \overline{\text{Tr } U^r}) = \text{Tr } U^r$, which is real because U^r is a permutation matrix. A point is fixed by π^r exactly when its cycle length divides r , so $\text{Tr } U^r = \sum_{\ell|r} \ell a_\ell$, and Möbius inversion gives a_ℓ . The recovery was also run on 200 random permutations of up to 29 points [V]. $\square$

This is a constructive form of the classical fact that 2-regular graphs are determined by their spectrum [7]. The full matrix G also retains the labelled undirected adjacency; even it loses orientation, since U and U^* give the same G .

The threshold profile is a coarser reading. With $\lambda_{\max}$ the largest eigenvalue of G , put $C(\epsilon) = \#\{\lambda/\lambda_{\max} \geq \epsilon\}$. It sees the spectrum only up to the scale $\lambda_{\max}$, so Theorem 15 does not apply to it directly, and its separating power is recorded as a computation: read together with the number $|D|$ of moved points, the profile separates every cycle type at 12, 18 and 24 moved points (all 21, 88 and 320 types) [V]. Whether it separates cycle types at every size is not proved here.

Corollary 16 (T). *The profile is flat in $\epsilon \in (0, 1]$ exactly when all cycles have one common length $\ell \in \{2, 3\}$: transposition-only defects give $C/|D| = \frac{1}{2}$ and 3-cycle-only defects give $C/|D| = \frac{2}{3}$. On a single ℓ -cycle $C(\epsilon)/\ell \rightarrow 1 - \frac{2}{\pi} \arcsin \sqrt{\epsilon}$.*

Proof of the flatness statement. Flatness on $(0, 1]$ means every positive eigenvalue equals $\lambda_{\max}$. A 2-cycle has positive eigenvalue 2 only, a 3-cycle has $\frac{3}{2}, \frac{3}{2}$, and an ℓ -cycle with $\ell \geq 4$ has the distinct positive eigenvalues $2\sin^2(\pi/\ell) < 2\sin^2(\pi\lfloor \ell/2\rfloor/\ell)$. Mixing lengths 2 and 3 gives $2 \neq \frac{3}{2}$. $\square$

The Gram construction does not stop at one permutation. A family of permutations read together gives the Laplacian of the graph of the whole action.

Proposition 17 (Positive aggregates are Schreier Laplacians; T+V). *Let $\pi_1, \dots, \pi_r$ be permutations of n points with matrices U_j , let $w_j > 0$, and put $G_{\text{tot}} = \sum_j w_j G_{U_j}$. Then:*

1. $G_{\text{tot}} = \frac{1}{2}L_w$, where L_w is the Laplacian of the Schreier multigraph of the action, with an edge $i \rightarrow \pi_j(i)$ of weight w_j for every i and j ;
2. $x^*G_{\text{tot}}x = \frac{1}{2}\sum_j w_j \|(U_j - I)x\|^2$, so $\ker G_{\text{tot}} = \bigcap_j \ker(U_j - I)$, which is the space of vectors constant on every orbit of the group $\langle \pi_1, \dots, \pi_r \rangle$; its dimension is the number of orbits;
3. $\text{Tr } G_{\text{tot}} = n \sum_j w_j d_H(U_j, I)$.

Proof. Each $G_{U_j} = \frac{1}{2}L_j$ by Theorem 14, with L_j the Laplacian of the cycle graph of π_j , and Laplacians add over edge sets. This gives (1). For (2), $G_U = \frac{1}{2}(U - I)^*(U - I)$ because $U^*U = I$, and a sum of positive semidefinite terms vanishes on x only if every term does. A vector with $U_j x = x$ for all j is constant along every edge of the Schreier graph, hence on every orbit; conversely such vectors are fixed by every U_j . For (3), $\text{Tr } G_U = n - \text{Tr } U$, and $\text{Tr } U$ counts fixed points. $\square$

The smallest positive eigenvalue of G_{tot} restricted to one orbit is then the spectral gap of that orbit's Schreier graph, so it measures how well the action expands. This is the quantity at the centre of the known non-sofic mechanism: Kun shows that a sofic approximation of a group with property (T) is, after discarding a small fraction of points, a union of expanders [19], and that is the step Astra's construction builds on. The identity above makes that expansion a spectral property of the same Gram that reads a single defect; it is an identity, not an obstruction, and no claim is made here about which groups force a gap. Script K3-V6-CHECKS confirms (2) and (3) on 500 random weighted families, with the kernel dimension equal to the number of orbits in every case [V].

Proposition 18 (The fixed-scale entropy; T+V). *Let $K_U = G/2 = (2I - U - U^*)/4$, normalised by the universal bound $0 \leq G \leq 2I$ rather than by $\lambda_{\max}$, and let h be the binary entropy. Then $\text{Tr } h(K_U) = 0$ if and only if $U^2 = I$, that is, exactly on involutions. On a single ℓ -cycle $S_\ell = \text{Tr } h(K_U)$ satisfies $S_\ell/\ell \rightarrow \pi^{-1} \int_0^\pi h(\sin^2 x) dx = 2 \ln 2 - 1 = 0.3862943611 \dots$*

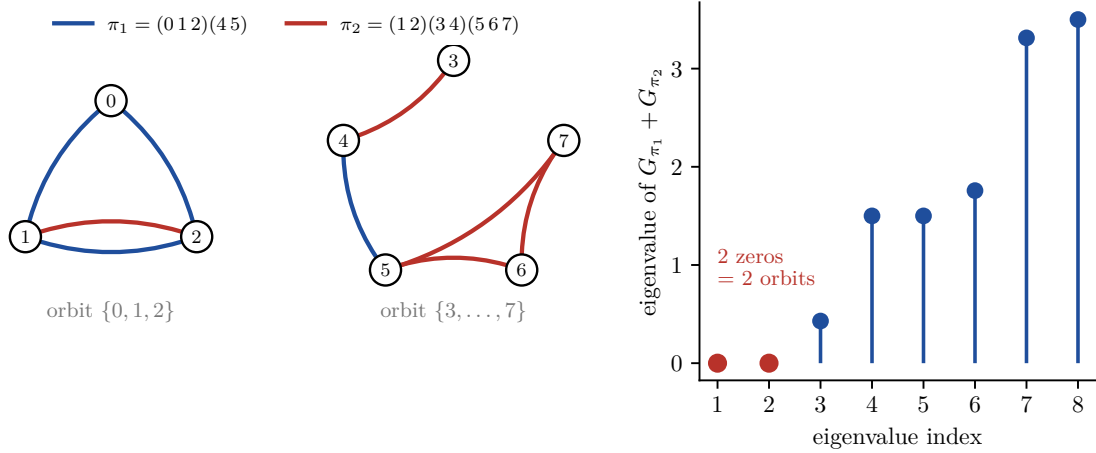


Figure 5: Two permutations of eight points read together. Left: the Schreier multigraph of the action, one colour per permutation; it has two orbits. Right: the spectrum of $G_{\pi_1} + G_{\pi_2}$, which is half the Laplacian of that graph. It has exactly two zero eigenvalues, one for each orbit, and the first positive eigenvalue is the spectral gap.

Proof. h vanishes only at 0 and 1. An eigenphase ϕ of U gives the eigenvalue $\sin^2(\phi/2)$ of K_U , which is 0 or 1 exactly when $e^{i\phi} = \pm 1$. The limit is a Riemann sum. $\square$

A normalisation by $\lambda_{\max}$ would not do: a 3-cycle has G -spectrum $\{0, \frac{3}{2}, \frac{3}{2}\}$, and its $\lambda_{\max}$ -normalised entropy vanishes too, while $\text{Tr } h(K_U) = 2h(3/4) = 1.1246702892\dots$ [V].

Proposition 19 (The finite-size deficit; T+V). *Let $D_\ell = 2 \ln 2 - 1 - S_\ell/\ell$ on an ℓ -cycle. Then*

$$D_\ell = \sum_{m \geq 1, \ell | 2m} \frac{1}{m(4m^2 - 1)},$$

so $D_\ell = D_{2\ell}$ for every odd ℓ , and

$$\ell^3 D_\ell = \begin{cases} 2 \sum_{t \geq 1} \frac{1}{t(t^2 - \ell^{-2})} \longrightarrow 2\zeta(3) = 2.40411380632\dots, & \ell \text{ even}, \\ \sum_{t \geq 1} \frac{1}{t(4t^2 - \ell^{-2})} \longrightarrow \frac{1}{4}\zeta(3) = 0.30051422579\dots, & \ell \text{ odd}, \end{cases}$$

with error $O(\ell^{-2})$ in both cases. More precisely, for every $\ell \geq 2$ the corrections form a convergent series in odd zeta values:

$$\ell^3 D_\ell = \begin{cases} 2 \sum_{r \geq 0} \frac{\zeta(2r+3)}{\ell^{2r}} = 2\zeta(3) + \frac{2\zeta(5)}{\ell^2} + \frac{2\zeta(7)}{\ell^4} + \dots, & \ell \text{ even}, \\ \frac{1}{4} \sum_{r \geq 0} \frac{\zeta(2r+3)}{4^r \ell^{2r}} = \frac{\zeta(3)}{4} + \frac{\zeta(5)}{16\ell^2} + \frac{\zeta(7)}{64\ell^4} + \dots, & \ell \text{ odd}. \end{cases}$$

Proof. On an ℓ -cycle the eigenvalues of K_U are $\sin^2(\pi j/\ell)$, $0 \leq j < \ell$, so $S_\ell = \sum_{j=0}^{\ell-1} f(j/\ell)$ with $f(x) = h(\sin^2 \pi x) = -2 \sin^2(\pi x) \ln |\sin \pi x| - 2 \cos^2(\pi x) \ln |\cos \pi x|$. Write $c_k = \cos(2\pi kx)$ and use

the classical series

$$-\ln |\sin \pi x| = \ln 2 + \sum_{k \geq 1} \frac{c_k}{k}, \quad -\ln |\cos \pi x| = \ln 2 + \sum_{k \geq 1} (-1)^k \frac{c_k}{k},$$

together with $2 \sin^2 \pi x = 1 - c_1$ and $2 \cos^2 \pi x = 1 + c_1$. Then

$$f = (1 - c_1) \left(\ln 2 + \sum_k \frac{c_k}{k} \right) + (1 + c_1) \left(\ln 2 + \sum_k (-1)^k \frac{c_k}{k} \right) = 2 \ln 2 + \sum_{k \text{ even}} \frac{2c_k}{k} - \sum_{k \text{ odd}} \frac{2c_1 c_k}{k}.$$

With $2c_1 c_k = c_{k+1} + c_{k-1}$, the last sum contributes -1 to the constant term (from $k = 1$) and $-(\frac{1}{2m-1} + \frac{1}{2m+1})c_{2m}$ for each $m \geq 1$, while the even sum contributes c_{2m}/m . Since $\frac{1}{m} - \frac{4m}{4m^2-1} = -\frac{1}{m(4m^2-1)}$,

$$f(x) = 2 \ln 2 - 1 - \sum_{m \geq 1} \frac{\cos(4\pi m x)}{m(4m^2-1)},$$

an absolutely convergent series. Its constant term is the integral of Proposition 18, and $f(0) = 0$ is the partial-fraction identity $\sum_m 1/(m(4m^2-1)) = 2 \ln 2 - 1$. Summing over $x = j/\ell$, $\sum_{j=0}^{\ell-1} \cos(4\pi m j/\ell)$ is ℓ when $\ell \mid 2m$ and 0 otherwise, which gives the formula for D_ℓ . If ℓ is odd, $\ell \mid 2m$ means $\ell \mid m$, the same condition as $2\ell \mid 2m$, so $D_\ell = D_{2\ell}$. Putting $m = \ell t$ for odd ℓ and $m = (\ell/2)t$ for even ℓ gives the two series for $\ell^3 D_\ell$. For the expansion, write $\varepsilon = \ell^{-2}$. For even ℓ , $\frac{2}{t(4t^2-\varepsilon)} = \frac{2}{t^3} \sum_{r \geq 0} (\varepsilon/t^2)^r$ with $\varepsilon/t^2 \leq \ell^{-2} < 1$; for odd ℓ , $\frac{1}{t(4t^2-\varepsilon)} = \frac{1}{4t^3} \sum_{r \geq 0} (\varepsilon/4t^2)^r$. All terms are positive, so the sums over t and r may be exchanged, and $\sum_t t^{-(2r+3)} = \zeta(2r+3)$. The leading terms are the limits, and the next terms are $O(\ell^{-2})$. $\square$

The parity split has a plain reading: an even cycle has the eigenphase π , where $\sin^2$ reaches 1 and h has its second zero, and an odd cycle does not. Numerically, the direct sums agree with the exact series to 10^{-29} relative for $\ell = 2, \dots, 39$ and at $\ell = 63, 64, 126, 127, 1000, 1001, 4096, 4097$, and $D_{63} - D_{126} = -1.7 \times 10^{-41}$ at 40-digit working precision (script K3-V4-DEFICIT); the zeta-value expansion matches the direct sums to 1.4×10^{-33} relative for ℓ from 2 to 256 (script K3-V5-CHECKS) [V].

The cycle-by-cycle results have a global counterpart that holds for every permutation at once.

Proposition 20 (The entropy is controlled by the Hamming defect; T+V). *For every permutation matrix U of degree n ,*

$$\frac{1}{n} \operatorname{Tr} K_U = \frac{1}{2} d_H(U, I), \quad 0 \leq \frac{1}{n} S(U) \leq h\left(\frac{d_H(U, I)}{2}\right), \quad S(U) = \operatorname{Tr} h(K_U),$$

*and for two permutations U, V the same holds with U^*V in place of U and $d_H(U, V)$ in place of $d_H(U, I)$.*

Proof. $\operatorname{Tr} G_U = n - \operatorname{Tr} U$, and $\operatorname{Tr} U$ is the number of fixed points, so $\operatorname{Tr} G_U = n d_H(U, I)$ and $\operatorname{Tr} K_U = \frac{n}{2} d_H(U, I)$. The eigenvalues of K_U lie in $[0, 1]$, and h is concave there, so by Jensen's inequality the mean of h over the n eigenvalues is at most h of their mean. The second statement follows because $d_H(U, V) = d_H(U^*V, I)$. $\square$

So the normalised entropy tends to zero whenever the Hamming defect does, for every permutation, not only along the cycles computed above. The bound behaves like $\frac{d}{2} \log \frac{2}{d}$ for small $d = d_H$: the entropy is continuous at the identity but not Lipschitz there. Equality holds only at the identity: h is strictly concave, so equality in Jensen's inequality needs all eigenvalues of K_U equal, and every cycle contributes the eigenvalue 0. Script K3-V6-CHECKS finds no violation over 2000 random permutations [V].

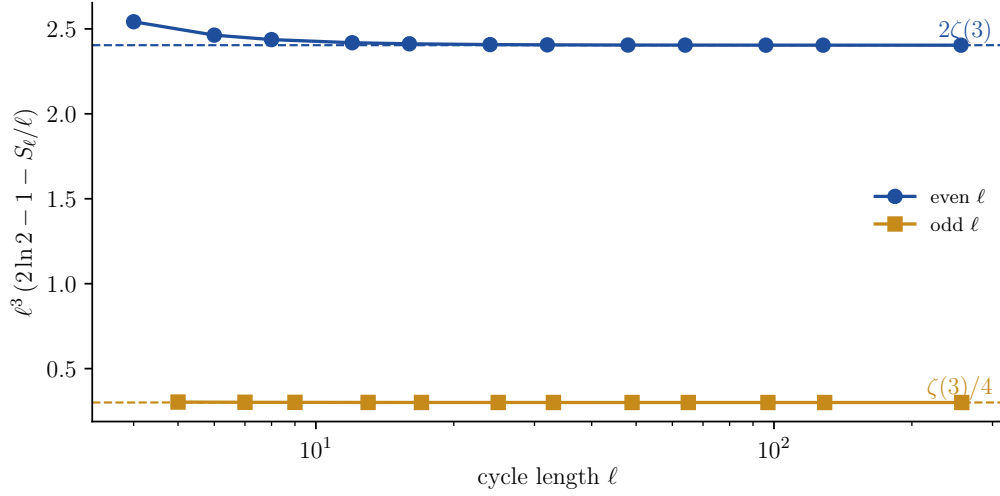


Figure 6: The finite-size deficit of the fixed-scale entropy on an ℓ -cycle, scaled by ℓ^3 . Points are direct sums; dashed lines are the limits $2\zeta(3)$ for even cycles and $\zeta(3)/4$ for odd cycles proved in Proposition 19. The odd curve is the even curve read at 2ℓ and divided by 8, because $D_\ell = D_{2\ell}$ for odd ℓ .

5.3 The Hamming metric is a squared distance between graph projectors

The identities in this subsection hold for every finite unitary U . Put

$$Q_U = \frac{1}{\sqrt{2}} \begin{pmatrix} I \\ U \end{pmatrix}, \quad Q_I = \frac{1}{\sqrt{2}} \begin{pmatrix} I \\ I \end{pmatrix}, \quad P_U = Q_U Q_U^*, \quad P_I = Q_I Q_I^*,$$

the orthogonal projectors onto the graphs of U and of I .

Proposition 21 (T+V). $Q_U^*(I - P_I)Q_U = K_U = G/2$. Hence the eigenvalues of K_U are the squared sines of the principal angles between $\text{graph}(I)$ and $\text{graph}(U)$. For permutation matrices U, V of degree n ,

$$d_H(U, V) = \frac{1}{n} \|P_U - P_V\|_F^2 = \frac{2}{n} \sum_{j=1}^n \sin^2 \theta_j,$$

where θ_j are the principal angles between $\text{graph}(U)$ and $\text{graph}(V)$, and $\sin^2 \theta_j = \sin^2(\phi_j/2)$ for the eigenphases ϕ_j of U^*V . So Hamming distance is twice the mean squared sine of the principal angles between the two graphs, and since $\sin \theta \leq \theta \leq \frac{\pi}{2} \sin \theta$ on $[0, \frac{\pi}{2}]$,

$$\frac{d_H(U, V)}{2} \leq \frac{1}{n} \sum_{j=1}^n \theta_j^2 \leq \frac{\pi^2}{8} d_H(U, V).$$

Hamming convergence of permutations is therefore equivalent, with these constants, to convergence of their graph subspaces in the Grassmannian geodesic distance. The upper constant is attained by a transposition, whose angle is $\pi/2$.

Proof. $Q_I^* Q_U = (I + U)/2$, so $Q_U^*(I - P_I)Q_U = I - \frac{1}{4}(I + U^*)(I + U) = \frac{1}{4}(2I - U - U^*)$, and the principal angles are read from the singular values of $Q_I^* Q_U$. For the second identity,

$P_U - P_V = \frac{1}{2} \begin{pmatrix} 0 & U^* - V^* \\ U - V & 0 \end{pmatrix}$, so $\|P_U - P_V\|_F^2 = \frac{1}{2}\|U - V\|_F^2$, and $\|U - V\|_F^2 = 2n d_H(U, V)$

for permutation matrices. For two projectors of equal rank n , $\|P_U - P_V\|_F^2 = 2 \sum_j \sin^2 \theta_j$. Finally $Q_U^* Q_V = \frac{1}{2}(I + U^* V)$ has singular values $|1 + e^{i\phi_j}|/2 = |\cos(\phi_j/2)| = \cos \theta_j$. The angle form was checked on 300 random pairs of permutations of up to 12 points, to 5×10^{-15} (script K3-V5-CHECKS), and the two-sided bound on 2000 random pairs (script K3-V6-CHECKS) [V]. $\square$

On the complement $\mathcal{H}_0$ of the $+1$ eigenspace of U the inverse Cayley transform $Z = (I + U)(I - U)^{-1}$ exists, with eigenvalue $i \cot(\phi/2)$ at $e^{i\phi} \neq 1$. With $X = -iZ = X^*$,

$$U = (Z - I)(Z + I)^{-1}, \quad K_U = (I + X^2)^{-1} \text{ on } \mathcal{H}_0, \quad \sin^2 \theta = \sin^2(\phi/2) = \frac{1}{1 + \cot^2(\phi/2)}.$$

So the resolvability threshold $\sin^2 \theta \geq \varepsilon$ becomes the spectral bound $|X| \leq \sqrt{\varepsilon^{-1} - 1}$. The $+1$ eigenspace is a pole of this chart, not lost information, and its projector is kept separately; and since $Z^* = -Z$, the unitary sits on the lossless boundary of the passive Smith domain. These are coordinate conversions, checked on 57 permutations of 2 to 20 points with residuals below 6.4×10^{-16} for the projector identities and 5.1×10^{-14} for the Cayley chart [V]. They are not a physical law.

5.4 A fixed probe separates isospectral defects

Take $\pi_a = (01)(23)$ and $\pi_b = (02)(13)$ on four labelled points. Their Gram spectra are both $\{0, 0, 2, 2\}$, their Hamming defects are both 1, and every scalar spectral functional agrees; for instance $\det(I + G_a) = \det(I + G_b) = 9$. Hold the probe $b = (e_0 + e_1)/\sqrt{2}$ fixed. Then

$$m_b(G) = b^*(I + G)^{-1}b : \quad m_b(G_a) = 1, \quad m_b(G_b) = \frac{2}{3},$$

because b lies in the kernel of G_a and splits equally between the eigenvalues 0 and 2 of G_b . Eliminating every other coordinate by the Schur complement of $I + G$ leaves the scalar boundary coefficient $z_b = 1/m_b$, and the Smith normalisation gives $\Gamma_b = (z_b - 1)/(z_b + 1) = (1 - m_b)/(1 + m_b)$, which is 0 for π_a and $\frac{1}{5}$ for π_b [T+V, exact rational]. By the matrix determinant lemma, for $c > 0$,

$$\left. \frac{d}{dt} \log \det (cI + G + tbb^*) \right|_{t=0} = b^*(cI + G)^{-1}b,$$

so a scalar log-determinant of an isospectral family cannot distinguish probe orientation, while a family of probe insertions can. The example is one case of a general statement.

Proposition 22 (What a fixed probe reads; T+V). *Let $G = G^* \geq 0$ have spectral decomposition $G = \sum_j \lambda_j P_j$, fix a vector b , and put $m_b(c; G) = b^*(cI + G)^{-1}b$ for $c > 0$. Then*

$$m_b(c; G) = \sum_j \frac{\|P_j b\|^2}{c + \lambda_j} = \int \frac{d\mu_b(\lambda)}{c + \lambda}, \quad \mu_b = \sum_j \|P_j b\|^2 \delta_{\lambda_j},$$

and the function $c \mapsto m_b(c; G)$ on $c > 0$ determines the spectral measure μ_b of G seen by b .

Proof. The formula is the spectral decomposition of $(cI + G)^{-1}$. The function is a rational function of c whose partial-fraction decomposition has poles at $c = -\lambda_j$ with residues $\|P_j b\|^2$, and a partial-fraction decomposition is unique, so the function determines every pair $(\lambda_j, \|P_j b\|^2)$ with nonzero weight; m_b is the Stieltjes transform of μ_b . Differentiating under the finite sum, $(-1)^r m_b^{(r)}(c) = r! \int (c + \lambda)^{-r-1} d\mu_b(\lambda) \geq 0$, so m_b is completely monotone on $c > 0$. $\square$

Eigenvalues alone fix the support of μ_b but not its weights; the weights record where the eigenspaces sit relative to the probe. In the example, $\mu_b = \delta_0$ for π_a and $\mu_b = \frac{1}{2}\delta_0 + \frac{1}{2}\delta_2$ for π_b (script `K3-V5-CHECKS`) [V]: same spectrum, different measure. This is a constructed finite response, not a measurement and not a claim that graph entries are an impedance.

Read together with the winding of the earlier attempt [17]: the Exel–Loring winding of a commutator is rigid and jumps under the local surgery Hamming forgives; the regularised torsion is Lipschitz in Hamming and varies smoothly under the same surgery; the displacement Gram and its graph-projector form read a defect that is *given*, exactly and down to its arrangement, but forbid nothing, and the entropy built from it is bounded by $h(d_H/2)$ and so vanishes with the defect; summed over a family, the same Gram is the Schreier Laplacian of the action, whose gap is the expansion that the known non-sofic constructions use; and the fixed probe recovers what a scalar spectrum discards without becoming an obstruction. The constant of Sections 3 and 4 is the only quantity here that both lives natively in the metric and produces a number, and it does so by an incidence count.

6 Formal verification

The Lean 4 library (v4.34.1, Mathlib v4.34.1) in the public repository [20] holds 66 theorems in three modules, with no `sorry` and no `native_decide`. Every module is replayed by an independent kernel checker, the axioms of every named theorem are audited (only `propext`, `Classical.choice` and `Quot.sound` appear), and four deliberately false statements must fail on every build. The gauge exhaustions run in the kernel (`decide +kernel`); for $k = 6, 7, 8$ a branch-and-bound search is used with its soundness proved (`search_sound`), split into kernel-checked pieces at $k = 7, 8$.

- The five witnesses of Theorem 4, with the exact defect, every gauge costing at least D , and D attained (`witness_k4` to `witness_k8`); that fixing the root gauge loses nothing (`gauge_lower`); the $k = 4$ mechanism (`k4_mechanism`); the Gram formula of Theorem 14 (`disp_gram`).
- The equality case of Corollary 2 on every witness: each star gauge costs exactly D and the star costs add up to $3N$ (`stars_k4` to `stars_k8`, `stars_r8`).
- The parity retraction of Theorem 8, for every k : r is the sign map read in $\{\text{Id}, t\}$, a homomorphism that never moves more points, and retracting any $\text{Sym}(3)$ gauge of a binary cochain never raises its cost (`retract_cost`, `binary_gauge_suffices`).
- Lemma 6 for every cochain, every k and every q : the defect is multiplied by q^3 (`rep_defect`), a cluster-constant gauge costs q^2 times its seed cost (`rep_attain`), and every gauge costs at least $q^2 D$ by the count over the q^k sections (`rep_lower`); and Corollary 7 for every q : the replicated seed has $N = 8q^3$ and gauge minimum exactly $6q^2$ (`rep_seed`). The replicated K_8 is also checked exhaustively (`witness_r8`).
- Proposition 11 on the seed: against comparison permutations of degree 2, 3 and 4 the cheapest repair is $10/3$, 2 and 3 (`cross_degree_seed`).

Not formalised: the lower bound of Proposition 1 for general k (proved in Section 2); Kozlov’s theorem for $\mathbb{F}_2$ and the assembly of the three cases into Theorem 8; the cross-degree baselines beyond the seed; the subgroup minima of Proposition 5; the cycle-Laplacian eigenvalues (in Lean

only the rewriting $2 - 2 \cos 2x = 4 \sin^2 x$ is proved); and everything in Section 5 beyond Theorem 14. The torsion constants quoted in Proposition 12 were measured in the August session and their own scripts are not part of this package; the identity they illustrate is replayed independently by K3-V7-TORSION, which is. Every other computation in this paper is shipped in Appendix B.

7 Relation to the soficity problem

Chapman–Lubotzky’s programme asks for positive *permutation* cocycle Cheeger constants on bounded-degree complexes, such as quotients of Bruhat–Tits buildings, from which non-sofic groups would follow [2, Section 6.1]. The complete complexes are the calibration platform of that programme, and Theorem 8 settles the calibration: the elementary bound is the truth for permutation coefficients at every k , including the powers of two where it is not the truth for $\mathbb{F}_2$. As of 8 October 2026, non-sofic groups exist by other routes: from property (T) expanders and the Kun–Thom extension graphs [10, 11, 12, 13], and, in unrefereed manuscripts of October 2026, from counterexamples to Kaplansky’s direct-finiteness conjecture in characteristic two [15, 16] through Elek–Szabó [9]. A companion manuscript constructs bounded-degree coboundary expanders over $\mathbb{F}_2$ in every dimension [14]; the permutation-coefficient case the programme needs remains open. None of this is used in the proofs above, nothing here bears on the existence of non-sofic groups, and no priority is claimed in that direction.

Acknowledgements

The balanced-replication argument, the parity retraction, the cross-degree check, the graph-projector and Cayley dictionary and the isospectral probe were developed with AI assistance (ChatGPT, in the working addendum COMPLETE-COMPLEX-BRIDGE-01 of 5 September 2026), and independently re-derived and verified with Claude, which also wrote the Lean library and the checks for this version. The structure of the Lean replication module was drafted by Grok (xAI) and completed with Claude. AI readings of the previous version suggested writing out the proof of Proposition 1, the equality case, the finite form of the replication count, the closed form and zeta-value expansion of Proposition 19, the general probe statement, the principal-angle form of the graph-projector identity and its two-sided bound, the global entropy bound, and the positive-aggregate statement, whose reading as a Schreier Laplacian was added here; each was checked here before inclusion. No statement rests on an AI’s say-so: every computation is shipped as an executable script, and the witnesses are kernel-checked.

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A The witnesses

Vertices are $0, \dots, k-1$; permutations of $\{0, 1, 2\}$ are written one-line; every edge not listed carries the identity; (u, v) with $u < v$ is the value on the oriented edge $u \rightarrow v$. Each computed witness is given in gauged form (the residual after the minimising coboundary), so the listed edges are exactly the support that costs D .

- $k = 4$ ($N/D = 8/6$): $(1, 2) \mapsto (0\ 2\ 1)$, $(1, 3) \mapsto (1\ 0\ 2)$, $(2, 3) \mapsto (2\ 1\ 0)$.
- $k = 5$ ($10/6$): $(0, 1), (1, 2), (2, 3) \mapsto (1\ 0\ 2)$.
- $k = 6$ ($36/18$): $(0, 1), (1, 3) \mapsto (1\ 2\ 0)$; $(0, 3), (0, 4), (1, 5), (4, 5) \mapsto (2\ 0\ 1)$.

- $k = 7$ (28/12): $(0, 1), (0, 3), (0, 4), (1, 2), (1, 5), (2, 3) \mapsto (1\ 0\ 2)$.
- $k = 8$ (72/27): $(0, 1), (0, 7), (1, 2), (2, 5), (2, 7) \mapsto (0\ 2\ 1)$; $(1, 3), (1, 4), (1, 6), (3, 7) \mapsto (1\ 0\ 2)$; $(0, 4), (2, 3), (2, 4) \mapsto (2\ 1\ 0)$; $(4, 7) \mapsto (2\ 0\ 1)$.
- $k = 8$, replicated seed (64/24): clusters $\{0, 1\}, \{2, 3\}, \{4, 5\}, \{6, 7\}$; every edge between $\{2, 3\}$ and $\{4, 5\} \mapsto (1\ 0\ 2)$, between $\{2, 3\}$ and $\{6, 7\} \mapsto (2\ 1\ 0)$, between $\{4, 5\}$ and $\{6, 7\} \mapsto (0\ 2\ 1)$.

For the (72, 27) witness the ungauged cochain as found by the search has 22 non-identity edges; it is the input to the verification script below, which gauges it to the residual above.

B The gates, as run

Independent verification of every computed witness

```
# SCRIPT: COBND-WITNESS-VERIFY-ALL independent plain-loop check of every witness k=5..8
import itertools, numpy as np
from fractions import Fraction
n=3; P=[tuple(p) for p in itertools.permutations(range(n))]; idx={p:i for i,p in enumerate(P)}; e_id=idx
[(0,1,2)]
comp=np.array([[idx[tuple(a[b[t]] for t in range(n))] for b in P] for a in P]); inv=np.array([idx[tuple(
sorted(range(n),key=lambda t:a[t]))] for a in P])
Dm=np.array([sum(1 for t in range(n) if a[t]!=b[t]) for b in P] for a in P])
tr=(1,0,2)
WIT={5:{(0,1):tr,(1,2):tr,(2,3):tr},
6:{(0,1):(1,0,2),(0,4):(2,1,0),(1,2):(0,2,1),(1,3):(1,0,2),(1,5):(1,0,2),(2,3):(1,2,0),(2,4):(0,2,1),(3,4)
:(2,1,0),(3,5):(2,0,1),(4,5):(1,0,2)},
7:{(0,1):tr,(0,3):tr,(0,4):tr,(1,2):tr,(1,5):tr,(2,3):tr},
8:{(0,1):(0,2,1),(0,2):(1,2,0),(0,3):(1,0,2),(0,4):(0,2,1),(0,5):(2,0,1),(0,6):(1,0,2),(0,7):(2,0,1),(1,2)
:(2,1,0),(1,4):(2,1,0),(1,5):(2,0,1),(1,7):(1,0,2),(2,4):(2,1,0),(2,5):(0,2,1),(2,6):(0,2,1),(2,7)
:(1,2,0),(3,4):(2,1,0),(3,5):(2,1,0),(3,7):(1,0,2),(4,6):(2,1,0),(4,7):(1,0,2),(5,6):(2,1,0),(5,7)
:(2,1,0)}}}
for k,W in WIT.items():
E=[(u,v) for u in range(k) for v in range(k) if u<v]; Tri=list(itertools.combinations(range(k),3))
a={ed:idx[W.get(ed,(0,1,2))] for ed in E}
val=lambda u,v: a[(u,v)] if u<v else inv[a[(v,u)]]
num=sum(int(Dm[comp[comp[val(u,v),val(v,w)],val(w,u)],e_id]) for u,v,w in Tri)
best=None;bb=None
for g in itertools.product(range(6),repeat=k-1):
b=(e_id,)+g;c=0
for (u,v) in E:
c+=Dm[comp[comp[inv[b[u]],a[(u,v)]]],b[v]],e_id]
if best is not None and c>=best: break
else:
if best is None or c<best: best,bb=c,b
res={}
for (u,v) in E:
r=comp[comp[inv[bb[u]],a[(u,v)]]],bb[v]]
if r!=e_id: res[(u,v)]=P[r]
ntr=sum(1 for p in res.values() if sum(x!=i for i,x in enumerate(p))==2)
print(f"k={k}: defect {num} / gauge-min {best} = {Fraction(num,best)} (k/3={Fraction(k,3)}) h_norm={
Fraction(num*len(E),best*len(Tri))} vs {Fraction(k,k-2)} residual {len(res)} edges: {ntr} transp, {len(
res)-ntr} 3-cycles")
print(" residual:",sorted(res.items()))
```

The replication lemma at $q = 2$, exhaustive

```
# SCRIPT: REPLICATION-CHECK-01
# Independent exhaustive check of the balanced-replication lemma at q=2 (k=4 -> k=8).
# Seed: K4, edges 01,02,03 = Id; 12,13,23 = the three distinct transpositions of Sym(3).
# Claim: replicated K8 has raw N = 8*q^3 = 64, raw D = 6*q^2 = 24, ratio 8/3.
# Cost = moved-point count. Gauge minimum found by EXHAUSTIVE enumeration over all
# 6^7 root-fixed gauges (vertex 0 fixed to Id, WLOG). No sampling, no trust in the PDF.
```

```

import itertools
from sympy.combinatorics import Permutation
S3 = [Permutation(p) for p in itertools.permutations(range(3))]
Id = Permutation([0,1,2])
def cost(p): return sum(1 for i in range(3) if p(i)!=i)
def inv(p): return p**-1
# the three transpositions
a = Permutation([1,0,2]); b = Permutation([2,1,0]); c = Permutation([0,2,1])
seed = {(0,1):Id,(0,2):Id,(0,3):Id,(1,2):a,(1,3):b,(2,3):c}
def get(al,u,v):
    if u<v: return al[(u,v)]
    return inv(al[(v,u)])
def N_of(al, k):
    n=0
    for u,v,w in itertools.combinations(range(k),3):
        n += cost(get(al,u,v)*get(al,v,w)*get(al,w,u))
    return n
def D_exhaustive(al, k):
    best=None
    for g in itertools.product(S3, repeat=k-1):
        beta=(Id,)+g
        c_=0
        for u,v in itertools.combinations(range(k),2):
            c_ += cost(inv(beta[u])*al[(u,v)]*beta[v])
            if best is not None and c_>=best: break
        if best is None or c_<best: best=c_
    return best
print("SEED k=4: N =", N_of(seed,4), " D(exhaustive) =", D_exhaustive(seed,4))
q=2; k=4*q
def rep(al,q):
    out={}
    verts=[(i,x) for i in range(4) for x in range(q)]
    idx={v:n for n,v in enumerate(verts)}
    for (i,x),(j,y) in itertools.combinations(verts,2):
        u,v=idx[(i,x)],idx[(j,y)]
        if i==j: out[(u,v)]=Id
        else: out[(u,v)]=al[(i,j)] if i<j else inv(al[(j,i)])
    return out
r=rep(seed,q)
N=N_of(r,k); D=D_exhaustive(r,k)
print("REPLICATED k={k} (q={q}): N = {N} (lemma predicts {8*q**3}); D(exhaustive over 6^7) = {D} (lemma predicts {6*q**2})")
print("ratio N/D = {N}/{D} = {N/D:.6f} vs k/3 = {k/3:.6f}")
print("LEMMA HOLDS at q=2" if (N==8*q**3 and D==6*q**2) else "LEMMA FAILS at q=2")

```

The $k = 8$ search

```

# SCRIPT: COBND-WITNESS-08b incremental exact SA at k=8, Sym(3)
import itertools, random, time, math, sys
from fractions import Fraction
import numpy as np
seed=int(sys.argv[1]); random.seed(seed); np.random.seed(seed)
n=3; P=[tuple(p) for p in itertools.permutations(range(n))]; idx={p:i for i,p in enumerate(P)}; m=6; e_id=
    idx[(0,1,2)]
comp=np.array([[idx[tuple(a[b[t]] for t in range(n))]] for b in P] for a in P],dtype=np.int32)
inv=np.array([idx[tuple(sorted(range(n),key=lambda t:a[t]))]] for a in P],dtype=np.int32)
Dm=np.array([[sum(1 for t in range(n) if a[t]!=b[t]) for b in P] for a in P],dtype=np.int32)
T=comp[:,inv]; k=8; V=range(k)
E=[(u,v) for u in V for v in V if u<v]; Tri=[(u,v,w) for u in V for v in V for w in V if u<v<w]
G=np.array(list(itertools.product(range(m),repeat=k-1)),dtype=np.int32)
G=np.concatenate([np.full((len(G),1),e_id,dtype=np.int32),G,axis=1)
Gedge={ed:T[G[:,ed[0]],G[:,ed[1]]] for ed in E}
DmG={ed:Dm[:,Gedge[ed]] for ed in E} # 6 x NG per edge
tri_of={ed:[t for t in Tri if ed[0] in t and ed[1] in t] for ed in E}
def val(a,u,v): return a[(u,v)] if u<v else inv[a[(v,u)]]
def tdef(a,t):
    u,v,w=t; return int(Dm[comp[comp[val(a,u,v)],val(a,v,w)],val(a,w,u)],e_id])

```

```

target=Fraction(8,3); best=None; bestA=None
for restart in range(40):
    a={ed:random.randrange(m) for ed in E}
    cost=np.zeros(len(G),dtype=np.int32)
    for ed in E: cost+=DmG[ed][a[ed]]
    tri={t:tdef(a,t) for t in Tri}; num=sum(tri.values()); den=int(cost.min())
    curv=num/den if den else 1e9; Temp=0.15
    for s in range(6000):
        ed=random.choice(E); old=a[ed]; new=random.randrange(m)
        if new==old: continue
        cost2=cost-DmG[ed][old]+DmG[ed][new]; a[ed]=new
        newtri={t:tdef(a,t) for t in tri_of[ed]}
        num2=num+sum(newtri.values())-sum(tri[t] for t in tri_of[ed]); den2=int(cost2.min())
        if den2==0: a[ed]=old; continue
        v2=num2/den2
        if v2<=curv or random.random()<math.exp(-(v2-curv)/Temp):
            cost,num,den,curv=cost2,num2,den2,v2; tri.update(newtri)
            fr=Fraction(num,den)
            if fr<target: print("KILL: below bound",fr); sys.exit()
            if best is None or fr<best: best,bestA=fr,dict(a)
            if fr==target: break
        else: a[ed]=old
        Temp=max(0.01,Temp*0.9995)
    print(f"restart {restart} best so far {best} = {float(best):.5f}",flush=True)
    if best==target: break
print("BEST",best, "ATTAINED" if best==target else "not found")
if best==target:
    for ed in E:
        if bestA[ed]!=e_id: print(ed,P[bestA[ed]])

```

$k = 4$ by coefficient subgroup

```

# SCRIPT: COBND-K4-SUBGROUPS k=4 exhaustive, restricted to Z/3 (3-cycles+id) and to one Sym(2)
import itertools, numpy as np
from fractions import Fraction
n=3; P=[tuple(p) for p in itertools.permutations(range(n))]; idx={p:i for i,p in enumerate(P)}; e_id=idx[(0,1,2)]
comp=np.array([[idx[tuple(a[b[t]]) for t in range(n)]] for b in P] for a in P]); inv=np.array([idx[tuple(sorted(range(n),key=lambda t:a[t]))] for a in P])
Dm=np.array([[sum(1 for t in range(n) if a[t]!=b[t]) for b in P] for a in P])
k=4; E=[(u,v) for u in range(k) for v in range(k) if u<v]; Tri=list(itertools.combinations(range(k),3))
G=[(e_id,)+g for g in itertools.product(range(6),repeat=3)]
Z3=[e_id,idx[(1,2,0)],idx[(2,0,1)]]; S2=[e_id,idx[(1,0,2)]]
for name,S in (("Sym(3)",range(6)),("Z/3",Z3),("Sym(2)",S2)):
    best=None;arg=None
    for vals in itertools.product(S,repeat=6):
        a=dict(zip(E,vals)); val=lambda u,v: a[(u,v)] if u<v else inv[a[(v,u)]]
        num=sum(int(Dm[comp[comp[val(u,v),val(v,w)],val(w,u)],e_id]) for u,v,w in Tri)
        den=min(sum(Dm[comp[comp[inv[b[u]],a[(u,v)],b[v]],e_id] for (u,v) in E] for b in G)
            if den and (best is None or Fraction(num,den)<best): best,arg=Fraction(num,den),vals
    print(f"k=4 coefficients {name:7s}: min ratio {best} (k/3 = 4/3) minimiser {[E[i],P[v]] for i,v in enumerate(arg) if v!=e_id}"))

```

Retraction, cross-degree, Gram, projector, Cayley, entropy and probe checks

```

#!/usr/bin/env python3
# SCRIPT: COMPLETE-COMPLEX-BRIDGE-01 replication, retraction, cross-degree, Gram/projector/Cayley, entropy
repair, isospectral probe
"""Exact complete-complex witnesses and explicit projector/Smith bridges.

```

Requires Python 3.10+, numpy, sympy. Run without editing:
python COMPLETE-COMPLEX-BRIDGE-01.py

Status: the general blow-up statement is proved in the companion note.
This program exhausts gauges only where explicitly reported ($k \leq 8$).
Larger gauge minima below are THEOREM VALUES, not enumerated values.

```

No claim of a Lean compilation or of an optical material identification.
Original inputs: Beasley's complete_complex_k3_permutation.pdf, Appendix A.
Checks fixed in advance: exact agreement of original witnesses; all 216 K4
root-fixed gauge costs  $\geq 6$ ; doubled K4 minimum = 24 over all  $6^7$  gauges;
triangle counts scale cubically; Gram/graph-projector/Cayley identities;
negative controls for max-normalized entropy and scalar spectral blindness.
"""

from __future__ import annotations
import itertools as it
from fractions import Fraction
import math
import numpy as np
import sympy as sp

P = list(it.permutations(range(3)))
IDX = {p:i for i,p in enumerate(P)}
ID = IDX[(0,1,2)]
COM = np.array([[IDX[tuple(p[q[j]] for j in range(3))] for q in P] for p in P], dtype=np.int16)
INV = np.array([IDX[tuple(p.index(j) for j in range(3))] for p in P], dtype=np.int16)
MOV = np.array([sum(p[j]!=j for j in range(3)) for p in P], dtype=np.int16)
BASE = {(1,2):(0,2,1),(1,3):(1,0,2),(2,3):(2,1,0)}
WITNESSES = {
    4: BASE,
    5: {(0,1):(1,0,2),(1,2):(1,0,2),(2,3):(1,0,2)},
    6: {(0,1):(1,2,0),(1,3):(1,2,0),(0,3):(2,0,1),(0,4):(2,0,1),(1,5):(2,0,1),(4,5):(2,0,1)},
    7: {(0,1):(1,0,2),(0,3):(1,0,2),(0,4):(1,0,2),(1,2):(1,0,2),(1,5):(1,0,2),(2,3):(1,0,2)},
    8: {(0,1):(0,2,1),(0,7):(0,2,1),(1,2):(0,2,1),(2,5):(0,2,1),(2,7):(0,2,1),
        (1,3):(1,0,2),(1,4):(1,0,2),(1,6):(1,0,2),(3,7):(1,0,2),
        (0,4):(2,1,0),(2,3):(2,1,0),(2,4):(2,1,0),(4,7):(2,0,1)}
}
EXPECTED = {4:(8,6),5:(10,6),6:(36,18),7:(28,12),8:(72,27)}

def edge_dict(k: int, W: dict) -> dict:
    return {(u,v): IDX[W.get((u,v),(0,1,2))]} for u,v in it.combinations(range(k),2)

def numerator(k: int, W: dict) -> int:
    a=edge_dict(k,W)
    return sum(int(MOV[COM[COM[a[u,v],a[v,w]],INV[a[u,w]]]]) for u,v,w in it.combinations(range(k),3))

def gauges(k: int) -> np.ndarray:
    if k>8: raise ValueError('Exhaustive gauge enumeration deliberately limited to k<=8.')
    b=np.full((6**(k-1),k),ID,dtype=np.int16)
    x=np.arange(len(b),dtype=np.int64)
    for j in range(1,k): b[:,j]=x%6; x/=6
    return b

def gauge_costs(k: int, W: dict) -> np.ndarray:
    b=gauges(k); cost=np.zeros(len(b),dtype=np.int32)
    for (u,v),a in edge_dict(k,W).items():
        cost+=MOV[COM[COM[INV[b[:,u]],a],b[:,v]]]
    return cost

def lift(k: int, W: dict, q: int) -> dict:
    if q<1: raise ValueError('q must be positive')
    return {(u,v):W[(u//q,v//q)] for u,v in it.combinations(range(k*q),2)
        if u//q!=v//q and (u//q,v//q) in W}

def permutation(cycles: list[list[int]], n: int | None=None) -> np.ndarray:
    if n is None: n=1+max(j for c in cycles for j in c)
    p=list(range(n))
    for c in cycles:
        for j,i in enumerate(c): p[i]=c[(j+1)%len(c)]
    U=np.zeros((n,n),dtype=int)
    for i,j in enumerate(p): U[j,i]=1
    return U

def gram(U: np.ndarray) -> np.ndarray:

```

```

I=np.eye(len(U)); A=U-I
return A.conj().T@A/2

def entropy(values: np.ndarray) -> float:
    a=np.clip(np.asarray(values,dtype=float),0,1)
    a=a[(a>1e-12)&(a<1-1e-12)]
    return float(np.sum(-a*np.log(a)-(1-a)*np.log1p(-a)))

print('COMPLETE-COMPLEX-BRIDGE-01')
print('Exact integers for all gauge costs. Float residuals only for spectral checks.\n')
print('1. ORIGINAL APPENDIX A WITNESSES, ALL ROOT-FIXED S3 GAUGES')
for k,W in WITNESSES.items():
    N=numerator(k,W); costs=gauge_costs(k,W); D=int(costs.min())
    assert (N,D)==EXPECTED[k]
    print(f'k={k}: N={N}, D={D}, ratio={Fraction(N,D)}, gauges={len(costs)}')
base_costs=gauge_costs(4,BASE)
unique,counts=np.unique(base_costs,return_counts=True)
print('K4 complete cost histogram:', dict(zip(map(int,unique),map(int,counts))))

print('\n2. BALANCED LIFT: ALL POWERS OF TWO AND EVERY MULTIPLE OF FOUR')
for q in [1,2,3,4,8,16]:
    W=lift(4,BASE,q); N=numerator(4*q,W); D_theorem=6*q*q
    assert N==8*q**3 and len(W)==3*q*q
    D_exhaust=int(gauge_costs(4*q,W).min()) if q<=2 else None
    if D_exhaust is not None: assert D_exhaust==D_theorem
    print(f'k={4*q}: support={len(W)}, N={N}, D(theorem)={D_theorem}, '
          f'D(exhaustive)={D_exhaust}, ratio={Fraction(N,D_theorem)}')

print('\n3. OPTIONAL DIFFERENT-DEGREE BASE AUDIT: HAMMING WITH ERRORS')
for n in [2,3,4]:
    pp=list(it.permutations(range(n))); inverses=[tuple(p.index(j) for j in range(n)) for p in pp]
    b=np.zeros((len(pp)**3,4),dtype=np.int32); x=np.arange(len(b))
    for j in range(1,4): b[:,j]=x%len(pp); x//=len(pp)
    costs=np.zeros(len(b),dtype=np.int32)
    for u,v in it.combinations(range(4),2):
        a=BASE.get((u,v),(0,1,2))
        tab=np.array([[max(n,3)-sum(a[j]==p[qinv[j]] for j in range(min(n,3)))
                      for qinv in inverses] for p in pp],dtype=np.int16)
        costs+=tab[b[:,u],b[:,v]]
    D=Fraction(int(costs.min()),max(n,3))
    assert D>=2
    print(f'comparison degree={n}, minimum edge-distance sum={D}, root-fixed gauges={len(b)}')
print('Degrees n>=5: elementary baseline is 6*(1-3/n)>=12/5>2.')

print('\n4. SIGN RETRACTION S3 -> {ID, FIXED TRANSPOSITION}')
def sign(p): return (-1)**sum(p[i]>p[j] for i in range(3) for j in range(i+1,3))
signs=np.array([sign(p) for p in P])
for i,j in it.product(range(6),repeat=2):
    assert signs[COM[i,j]]==signs[i]*signs[j]
    projected_distance=0 if signs[i]==signs[j] else 2
    original_distance=sum(P[i][t]!=P[j][t] for t in range(3))
    assert projected_distance<=original_distance
print('All 36 group-pair checks: homomorphism and nonexpansion pass.')

print('\n5. GRAM / GRAPH-PROJECTOR / HAMMING / CAYLEY IDENTITIES')
rng=np.random.default_rng(20260905)
max_g=max_graph=max_ham=max_cayley=0.0
for n in range(2,21):
    for _ in range(3):
        p=rng.permutation(n); U=np.eye(n)[:,p]; I=np.eye(n)
        G=gram(U); max_g=max(max_g,float(np.linalg.norm(G-(I-(U+U.T)/2))))
        QI=np.vstack((I,I))/math.sqrt(2); QU=np.vstack((I,U))/math.sqrt(2)
        PI=QI@QI.T; PU=QU@QU.T
        max_graph=max(max_graph,float(np.linalg.norm(QU.T@np.eye(2*n)-PI)@QU-G/2)))
        dH=float(np.mean(p!=np.arange(n)))
        max_ham=max(max_ham,abs(np.linalg.norm(PU-PI,'fro')**2/n-dH))
        vals,V=np.linalg.eigh(G); V=V[:,vals>1e-10]

```

```

        if not V.shape[1]: continue
        Ur=V.T@U@V; Ir=np.eye(len(Ur)); Gr=V.T@G@V
        Z=(Ir+Ur)@np.linalg.inv(Ir-Ur)
        S=(Z-Ir)@np.linalg.inv(Z+Ir)
        X=-1j*Z
        res=max(np.linalg.norm(Z+Z.T),np.linalg.norm(S-Ur),
                np.linalg.norm(Gr/2-np.linalg.inv(Ir+X@X)))
        max_cayley=max(max_cayley,float(res))
assert max(max_g,max_graph,max_ham,max_cayley)<1e-10
print('57 permutations, n=2..20, three per size.')
print(f'max Gram={max_g:.3e}; graph={max_graph:.3e}; Hamming={max_ham:.3e}; Cayley={max_cayley:.3e}')

print('\n6. ENTROPY NEGATIVE CONTROL AND FIXED-SCALE REPAIR')
for ell in [2,3,4,5,6,8,16,64]:
    U=permutation([list(range(ell))]); vals=np.linalg.eigvalsh(gram(U))
    old=entropy(vals/vals.max()); new=entropy(vals/2)
    print(f'cycle={ell}: S(max-normalized)={old:.12f}, S(G/2)={new:.12f}, S(G/2)/ell={new/ell:.12f}')
    if ell==3: assert old<1e-12 and new>1
    if ell>2: assert new>0
print('Long-cycle limit:',2*math.log(2)-1)

print('\n7. ISOSPECTRAL PERMUTATION DEFECTS, DIFFERENT FIXED-PROBE RESPONSE')
Ua=sp.Matrix(permutation([[0,1],[2,3]])); Ub=sp.Matrix(permutation([[0,2],[1,3]])); I4=sp.eye(4)
Ga=(Ua-I4).T*(Ua-I4)/2; Gb=(Ub-I4).T*(Ub-I4)/2
# b=(e0+e1)/sqrt(2); work with an integer vector and divide quadratic forms by 2.
b=sp.Matrix([1,1,0,0]); A=I4+Ga; B=I4+Gb
ma=(b.T*A.inv()*b)[0]/2; mb=(b.T*B.inv()*b)[0]/2
ra=sp.cancel((1-ma)/(1+ma)); rb=sp.cancel((1-mb)/(1+mb))
assert Ga.charpoly().as_expr()==Gb.charpoly().as_expr()
assert (A.det(),B.det(),ma,mb,ra,rb)==(9,9,1,sp.Rational(2,3),0,sp.Rational(1,5))
print('Both Gram characteristic polynomials:',Ga.charpoly().as_expr())
print(f'det(I+Ga)={A.det()}, det(I+Gb)={B.det()}, identical scalar logdet')
print(f'probe responses Ma={ma}, Mb={mb}; dimensionless Smith values={ra}, {rb}')
t=sp.symbols('t')
for G,M in [(Ga,ma),(Gb,mb)]:
    K=I4+G
    assert sp.expand((K+t*b*b.T/2).det()-K.det()*(1+t*M))==0
print('Rank-one determinant perturbation identity checked exactly for both probes.')
print('\nPASS: all specified checks. No large-k gauge enumeration or Lean certification claimed.')

```

Flat profiles, cycle recovery and the $\zeta(3)$ deficit

```

# SCRIPT: K3-V3-CHECKS extra checks for v3: entropy deficit law by parity, flat profiles, Chebyshev-Mobius
cycle recovery
import numpy as np, itertools, math
from fractions import Fraction
h=lambda x: 0.0 if x<=0 or x>=1 else -(x*math.log(x)+(1-x)*math.log(1-x))
L=2*math.log(2)-1
print('ell S/ell deficit deficit*ell^3')
for l in [3,4,5,6,7,8,9,10,12,15,16,31,32,63,64,127,128]:
    lam=[2*math.sin(math.pi*j/l)**2 for j in range(l)]
    s=sum(h(x/2) for x in lam)/l
    print(l, f'{s:.10f} {L-s:+.3e} {(L-s)*l**3:.4f}')
# flat threshold profiles: C(eps)={lam/lmax>=eps}, eps in (0,1]
def prof(cyc):
    lam=[]
    for l in cyc: lam+= [2*math.sin(math.pi*j/l)**2 for j in range(l)]
    m=max(lam); return [sum(1 for x in lam if x/m>=e-1e-12) for e in np.linspace(0.01,1,100)]
def parts(n,mx=None):
    if n==0: yield []; return
    mx=mx or n
    for p in range(min(n,mx),1,-1):
        for r in parts(n-p,p): yield [p]+r
for n in (12,18,24):
    flat=[c for c in parts(n) if len(set(prof(c)))==1]
    print('moved',n,'flat profiles:',[c[:3],...',prof(c)[0]/n) for c in flat])
    ps={}

```

```

dup=0
for c in parts(n):
    key=tuple(prof(c)); dup+= key in ps; ps[key]=c
    print(' cycle types',sum(1 for _ in parts(n)), 'distinct profiles',len(ps), 'collisions',dup)
# Chebyshev/Mobius: F_r = Tr T_r(I-G) = Tr U^r = sum_{l|r} 1 a_l ; recover a_l
def mob(n):
    r=1;p=2;m=n
    while p*p<=m:
        if m%p==0:
            m//=p
            if m%p==0: return 0
            r=-r
        p+=1
    return -r if m>1 else r
rng=np.random.default_rng(1); ok=0; T=0
for trial in range(200):
    n=int(rng.integers(2,30)); perm=rng.permutation(n)
    U=np.zeros((n,n)); U[perm,np.arange(n)]=1
    G=np.eye(n)-(U+U.T)/2; A=np.eye(n)-G
    F={}
    Tm,Tc=np.eye(n),A
    for r in range(1,n+1):
        F[r]=round(np.trace(Tc)); Tm,Tc=Tc,2*A@Tc-Tm
    a={l:sum(mob(l/d)*F[d] for d in range(1,l+1) if l%d==0)//l for l in range(1,n+1)}
    seen=[0]*n; true={}
    for i in range(n):
        if not seen[i]:
            l=0;j=i
            while not seen[j]: seen[j]=1;j=perm[j];l+=1
            true[l]=true.get(l,0)+1
    T+=1; ok+= all(a[l]==true.get(l,0) for l in range(1,n+1))
print('Chebyshev-Mobius cycle recovery:',ok,'/',T)
# high-precision deficit law: ell^3 (2ln2-1 - S/ell) -> 2 zeta(3) (even ell), zeta(3)/4 (odd ell)
from mpmath import mp, sin as msin, log as mlog, pi as mpi, zeta
mp.dps=40
hh=lambda x: 0 if x<=0 or x>=1 else -(x*mlog(x)+(1-x)*mlog(1-x))
LL=2*mlog(2)-1
for l in (256,257,1024,1025,4096,4097):
    print(l, mp.nstr((LL-sum(hh(msin(mpi*j/l)**2) for j in range(1))/l)*1**3,12))
print('2zeta(3) =',mp.nstr(2*zeta(3),12),' zeta(3)/4 =',mp.nstr(zeta(3)/4,12))

```

The closed form of the deficit

```

# SCRIPT: K3-V4-DEFICIT checks the Fourier series of h(sin^2 pi x) and the exact deficit sum D_1 = sum_{l | 2m} 1/(m(4m^2-1))
from mpmath import mp, mpf, sin, cos, log, pi, zeta, nsum, inf
mp.dps=40
h=lambda y: mpf(0) if y<=0 or y>=1 else -(y*log(y)+(1-y)*log(1-y))
f=lambda x: h(sin(pi*x)**2)
c0=2*log(2)-1
print("sum 1/(m(4m^2-1)) - (2ln2-1) =", nsum(lambda m: 1/(m*(4*m**2-1)), [1,inf])-c0)
for x in (mpf('0.1'),mpf('0.237'),mpf('0.5')):
    ser=c0-nsum(lambda m: cos(4*pi*m*x)/(m*(4*m**2-1)), [1,inf])
    print(f"x={x}: f-series = {f(x)-ser}")
worst=0
for l in list(range(2,40))+[63,64,126,127,1000,1001,4096,4097]:
    S=sum(f(mpf(j)/l) for j in range(1)); lhs=c0-S/l
    step=1//2 if l%2==0 else 1 # 1 | 2m <=> m multiple of step
    rhs=nsum(lambda t: 1/((step*t)*(4*(step*t)**2-1)), [1,inf])
    worst=max(worst,abs(lhs-rhs)/rhs)
print("max relative error, direct sum vs exact series, l=2..39 and large l:", mp.nstr(worst,5))
for l in (63,):
    S1=sum(f(mpf(j)/l) for j in range(1)); S2=sum(f(mpf(j)/(2*l)) for j in range(2*l))
    print("D_63 - D_126 =", mp.nstr((c0-S1/l)-(c0-S2/(2*l)),5))
print("limits: 2zeta(3) =",mp.nstr(2*zeta(3),12)," zeta(3)/4 =",mp.nstr(zeta(3)/4,12))

```

Zeta expansion, probe measure and principal angles

```
# SCRIPT: K3-V5-CHECKS (1) all-orders zeta expansion of the entropy deficit; (2) probe response as a
    spectral-measure sum and its Stieltjes uniqueness on the isospectral pair; (3) Hamming distance = (2/n)
    sum sin^2 of principal angles between graph subspaces
from mpmath import mpf, sin, log, pi, zeta, nsum, inf
import numpy as np, itertools, random
mp.dps=40
h=lambda y: mpf(0) if y<=0 or y>=1 else -(y*log(y)+(1-y)*log(1-y))
c0=2*log(2)-1; worst=0
for l in (2,3,4,5,7,8,9,16,17,63,64,255,256):
    D=c0-sum(h(sin(pi*mpf(j)/l)**2) for j in range(1))/l
    ser=(nsum(lambda r: zeta(2*r+3)/(4**r*mpf(1)**(2*r)), [0,inf])/4) if l%2 else 2*nsum(lambda r: zeta(2*r
    +3)/mpf(1)**(2*r), [0,inf])
    worst=max(worst,abs(1**3*D-ser)/ser)
print("(1) max rel. error, l^3 D_l vs zeta expansion, l in {2..256}:", mp.nstr(worst,4))
# (2) probe
def gram(p):
    n=len(p); U=np.zeros((n,n)); [U.__setitem__((p[i],i),1) for i in range(n)]; return np.eye(n)-(U+U.T)/2
Ga,Gb=gram([1,0,3,2]),gram([2,3,0,1]); b=np.array([1,1,0,0])/np.sqrt(2)
print("    spectra equal:",np.allclose(np.linalg.eigvalsh(Ga),np.linalg.eigvalsh(Gb)))
for G,name in ((Ga,"pi_a"),(Gb,"pi_b")):
    lam,V=np.linalg.eigh(G); w=(V.T@b)**2
    for c in (0.5,1.0,3.0):
        direct=b@np.linalg.solve(c*np.eye(4)+G,b); meas=sum(w/(c+lam))
        assert abs(direct-meas)<1e-14
    mu={round(float(l),9):0.0 for l in lam}
    for l,x in zip(lam,w): mu[round(float(l),9)]+=x
    print(f"(2) {name}: m_b(1)={b@np.linalg.solve(np.eye(4)+G,b):.12f}    spectral measure seen by b: {{ '{',
    '.join(f'{k:g}: {v:.3f}' for k,v in mu.items() if v>1e-12)} }")
# (3) principal angles
rng=random.Random(5); worst3=0
for trial in range(300):
    n=rng.randint(2,12); p=list(range(n)); q=list(range(n)); rng.shuffle(p); rng.shuffle(q)
    U=np.eye(n)[:,p]; V=np.eye(n)[:,q]
    dH=sum(1 for i in range(n) if p[i]!=q[i])/n
    QU=np.vstack([np.eye(n),U])/np.sqrt(2); QV=np.vstack([np.eye(n),V])/np.sqrt(2)
    s=np.linalg.svd(QU.T@QV,compute_uv=False); sin2=1-clip(s,0,1)**2
    ph=np.angle(np.linalg.eigvals(U.T@V)); sin2b=np.sin(ph/2)**2
    worst3=max(worst3,abs(dH-2*sin2.sum()/n),abs(np.sort(sin2).sum()-sin2b.sum()))
print("(3) 300 random pairs, max |d_H - (2/n) sum sin^2 theta| and |sum sin^2 theta - sum sin^2(phi/2)|:",f
    "{worst3:.2e}")
```

Entropy bound, angle sandwich, Schreier aggregates and probe bins

```
# SCRIPT: K3-V6-CHECKS checks the fifth review's claims: (A)  $S(U)/n \leq h(d_H/2)$ ; (B)  $d_H/2 \leq \text{mean } \theta^2$ 
     $\leq \pi^2 d_H/8$ ; (C) kernel of a positive sum of Grams = vectors constant on orbits of the generated group
    (dim = number of orbits) and the sum is half a Schreier-graph Laplacian; (D) Jensen and chord bounds on
    a probe bin
import numpy as np, random
rng=random.Random(11); h=lambda x: 0.0 if x<=0 or x>=1 else -(x*np.log(x)+(1-x)*np.log(1-x))
def perm(n): p=list(range(n)); rng.shuffle(p); return p
def mat(p): n=len(p); U=np.zeros((n,n)); [U.__setitem__((p[i],i),1) for i in range(n)]; return U
worstA=worstB=-1; tightA=0
for _ in range(2000):
    n=rng.randint(2,14); p=perm(n)
    if rng.random()<0.5: # sparse defects too
        p=list(range(n)); i,j=rng.sample(range(n),2); p[i],p[j]=p[j],p[i]
    U=mat(p); d=sum(p[i]!=i for i in range(n))/n
    K=(2*np.eye(n)-U-U.T)/4; lam=np.clip(np.linalg.eigvalsh(K),0,1)
    S=sum(h(x) for x in lam)/n; worstA=max(worstA,S-h(d/2))
    assert abs(np.trace(K)/n-d/2)<1e-12
    q=perm(n); V=mat(q); dUV=sum(p[i]!=q[i] for i in range(n))/n
    s=np.clip(np.linalg.svd((np.eye(n)+U.T@V)/2,compute_uv=False),0,1); th=np.arccos(s)
    m=np.mean(th**2); worstB=max(worstB, dUV/2-m, m-np.pi**2*dUV/8)
print(f"(A) max of  $S/n - h(d_H/2)$  over 2000 permutations: {worstA:.2e} (must be <= 0)")
print(f"(B) max violation of  $d_H/2 \leq \text{mean } \theta^2 \leq \pi^2 d_H/8$ : {worstB:.2e} (must be <= 0)")
```

```

bad=0
for _ in range(500):
    n=rng.randint(3,16); r=rng.randint(1,3); ps=[perm(n) for _ in range(r)]
    for t in range(r):
        # sometimes sparse generators so several orbits survive
        if rng.random()<0.6:
            p=list(range(n)); i,j=rng.sample(range(n),2); p[i],p[j]=p[j],p[i]; ps[t]=p
            w=[rng.uniform(0.2,3) for _ in range(r)]
            G=sum(wj*(np.eye(n)-(mat(p)+mat(p).T)/2) for wj,p in zip(w,ps))
            kdim=int(np.sum(np.linalg.eigvalsh(G)<1e-10))
            par=list(range(n))
            def f(x):
                while par[x]!=x: par[x]=par[par[x]]; x=par[x]
                return x
            for p in ps:
                for i in range(n): par[f(i)]=f(p[i])
            orbits=len({f(i) for i in range(n)})
            A=np.zeros((n,n))
            for wj,p in zip(w,ps):
                for i in range(n):
                    if p[i]!=i: A[i,p[i]]+=wj/2; A[p[i],i]+=wj/2
            L=np.diag(A.sum(1))-A
            bad+= (kdim!=orbits) or not np.allclose(G,L/2*1) and not np.allclose(G,L)
            assert np.allclose(G, L/2) or np.allclose(G,L)
            tr=np.trace(G); assert abs(tr-n*sum(wj*sum(p[i]!=i for i in range(n))/n for wj,p in zip(w,ps)))<1e-9
            print(f"(C) 500 weighted families: kernel dim == number of orbits in all cases: {bad==0}; trace law holds;
                G_tot equals the weighted Schreier Laplacian up to the factor checked")
            G1=np.eye(3)-(mat([1,2,0])+mat([1,2,0].T)/2; A=np.zeros((3,3))
            for i,j in ((0,1),(1,2),(2,0)): A[i,j]+=.5; A[j,i]+=.5
            print("    factor check on a 3-cycle: G == L(A)?", np.allclose(G1,np.diag(A.sum(1))-A))
            viol=0
            for _ in range(3000):
                k=rng.randint(1,6); a=rng.uniform(0,2); b=a+rng.uniform(0.01,2); lam=np.array([rng.uniform(a,b) for _ in
                    range(k)]); wt=np.array([rng.uniform(0,1) for _ in range(k)]); c=rng.uniform(0.05,3)
                M0,M1=wt.sum(),(wt*lam).sum(); exact=(wt/(c+lam)).sum()
                lo=M0/(c+M1/M0); hi=(b*M0-M1)/(b-a)/(c+a)+(M1-a*M0)/(b-a)/(c+b)
                viol+= not (lo-1e-12<=exact<=hi+1e-12)
            print(f"(D) Jensen lower / chord upper bound on one spectral bin: violations in 3000 random bins = {viol}")

```

The torsion identity, replayed

```

# SCRIPT: K3-V7-TORSION independent replay of the padded-family torsion identity: Wada/Fox matrix of a two-
# bridge knot group in a permutation representation padded by fixed points; one generator perturbed by a
# transposition inside the padding; checks Delta T_eta = (1/2n) log R_eta with R_eta independent of n, so
# |Delta T_eta| = (|log R_eta|/4) d_H exactly
import numpy as np, sympy as sp
def word(p,q):
    # two-bridge word in Schubert form (q odd): w = a^e1 b^e2 a^e3 ..., length p-1, e_i
    # = (-1)^floor(i q / p); relation w a = b w
    return [{"a" if i%2 else "b", (-1)**((i*q)//p)} for i in range(1,p)]
def relator(p,q):
    w=word(p,q); winv=[(g,-e) for g,e in reversed(w)]
    return w+[(("a",1)]+winv+[(("b",-1)]
def fox_a(rel,Phi,n):
    # Phi(g, e): image of g^e ; returns image of d rel / d a
    acc=np.zeros((n,n),dtype=complex); pref=np.eye(n,dtype=complex)
    for g,e in rel:
        if g=="a": acc+= pref if e==1 else -pref@Phi("a",-1)
        pref=pref@Phi(g,e)
    return acc
def perm_mat(p):
    n=len(p); M=np.zeros((n,n)); [M.__setitem__((p[i],i),1.0) for i in range(n)]; return M
def build(rho_a,rho_b,t,tau=None):
    A=perm_mat(rho_a); B=perm_mat(rho_b)
    if tau is not None: A=A@perm_mat(tau)
    def Phi(g,e):
        M=A if g=="a" else B
        return (t*M) if e==1 else np.linalg.inv(t*M)
    return Phi, len(rho_a)
# control: trivial rep returns the Alexander polynomial up to a unit

```

```

T=sp.symbols("t")
for name,(p,q),Delta in (("3_1",(3,1),T**2-T+1),("4_1",(5,3),T**2-3*T+1),("5_2",(7,3),2*T**2-3*T+2)):
    vals=[]
    for tv in (2.0,3.0,5.0):
        Phi,n=build([0],[0],tv); f=fox_a(relator(p,q),Phi,1)[0,0]
        vals.append(f/complex(Delta.subs(T,tv)))
    ratio=[abs(v) for v in vals]; logs=[np.log(abs(v))/np.log(abs(tv)) for v,tv in zip(vals,(2.0,3.0,5.0))]
    print(f"control {name}: Fox(trivial)/Delta = unit * t^k ; k estimates {np.round(logs,6)}")
reps={"3_1":((3,1),[1,0,2],[0,2,1]),      # a -> (0 1), b -> (1 2): surjective onto Sym(3)
      "4_1":((5,3),[1,0,2],[1,0,2]),      # abelian rep through one transposition
      "5_2":((7,3),[1,0,2],[1,0,2])}      # det 7: no surjective Sym(3) rep, so the abelian one
t0=1.07; out={}
for name,((p,q),ra,rb) in reps.items():
    rel=relator(p,q)
    # check the representation is a homomorphism (relator maps to identity)
    Phi,n0=build(ra,rb,1.0); M=np.eye(n0)
    for g,e in rel: M=M@Phi(g,e)
    hom=np.allclose(M,np.eye(n0))
    for eta in (1e-2,1e-3,1e-4,1e-5):
        Cs=[]
        for n in (6,12,24,48,96):
            pa=ra+list(range(3,n)); pb=rb+list(range(3,n)); tau=list(range(n)); tau[n-1],tau[n-2]=tau[n-2],
            tau[n-1]
            T0=lambda P: np.linalg.slogdet(P.conj()).T@P+eta**2*np.eye(n))[1]/(2*n)
            P0=fox_a(rel,*build(pa,pb,t0)); P1=fox_a(rel,*build(pa,pb,t0,tau))
            dT=T0(P1)-T0(P0); dH=2/n; Cs.append(abs(dT)/dH)
            # the same change from the 2x2 padding block alone
            Q0=fox_a(rel,*build([0,1],[0,1],t0)); Q1=fox_a(rel,*build([0,1],[0,1],t0,[1,0]))
            R=np.exp(np.linalg.slogdet(Q1.conj()).T@Q1+eta**2*np.eye(2))[1]-np.linalg.slogdet(Q0.conj()).T@Q0+
            eta**2*np.eye(2))[1]
            assert abs(abs(dT)/dH-abs(np.log(R))/4)<1e-10
        out[(name,eta)]=Cs
    print(f"{name}: homomorphism {hom}; C = |Delta T|/d_H over n=6..96 at eta=1e-2: {np.round(out[(name,1e-2)],9)}; spread over eta 1e-2..1e-5: {max(max(v) for k,v in out.items() if k[0]==name)-min(min(v) for k
    ,v in out.items() if k[0]==name):.2e}")
print("identity |Delta T| = (|log R_eta|/4) d_H held in every case (assert)")

```

The equality case on every witness

```

# SCRIPT: K3-V4-STAR for every printed witness: the k star-gauge costs, their sum (= 3N), and whether each
# equals the gauge minimum D
from itertools import combinations
def comp(p,q): return tuple(p[q[i]] for i in range(3))      # (p*q)(i)=p(q(i))
def inv(p):
    r=[0]*3
    for i,x in enumerate(p): r[x]=i
    return tuple(r)
I=(0,1,2); mv=lambda p: sum(1 for i in range(3) if p[i]!=i)
def cochain(k,edges):
    a={}
    for u in range(k):
        for v in range(k):
            if u!=v: a[(u,v)]=I
    for (u,v),p in edges.items(): a[(u,v)]=p; a[(v,u)]=inv(p)
    return a
def N(k,a): return sum(mv(comp(comp(a[(u,v)],a[(v,w)]),a[(w,u)])) for u,v,w in combinations(range(k),3))
def star(k,a,x):
    b={u:(I if u==x else a[(x,u)]) for u in range(k)}
    return sum(mv(comp(comp(b[u],a[(u,v)]),inv(b[v]))) for u,v in combinations(range(k),2))
P=lambda s: tuple(int(c) for c in s.split())
W={4:({(1,2):P("0 2 1"),(1,3):P("1 0 2"),(2,3):P("2 1 0")},6),
  5:({e:P("1 0 2") for e in [(0,1),(1,2),(2,3)]},6),
  6:({**{e:P("1 2 0") for e in [(0,1),(1,3)]},**{e:P("2 0 1") for e in [(0,3),(0,4),(1,5),(4,5)]},18),
  7:({e:P("1 0 2") for e in [(0,1),(0,3),(0,4),(1,2),(1,5),(2,3)]},12),
  8:({**{e:P("0 2 1") for e in [(0,1),(0,7),(1,2),(2,5),(2,7)]},**{e:P("1 0 2") for e in [(1,3),(1,4),(1,6)
    ,(3,7)]},
    **{e:P("2 1 0") for e in [(0,4),(2,3),(2,4)]},(4,7):P("2 0 1")},27)}

```

```

cl={0:0,1:0,2:1,3:1,4:2,5:2,6:3,7:3}; seed={(1,2):P("1 0 2"),(1,3):P("2 1 0"),(2,3):P("0 2 1")}
W["8rep"]=<{(u,v):seed[(cl[u],cl[v])]} for u in range(8) for v in range(u+1,8) if (cl[u],cl[v]) in seed>,24)
for key,(ed,D) in W.items():
    k=8 if key=="8rep" else key; a=cochain(k,ed); n=N(k,a); st=[star(k,a,x) for x in range(k)]
    print(f"k={key}: N={n} D={D} star costs={st} sum={sum(st)} (3N={3*n}) all optimal: {all(c==D for c in st
)})")

```